Test Run Report — Run #108

Environment: **Production** • Trigger: Manual • Started: Aug 5, 2026 04:53 • Ended: Aug 5, 2026 07:43

Generated: 8/5/2026, 8:12:19 AM

Summary

210	89.9%	170 / 19	21
Total Tests	Pass Rate (excl. skipped)	Passed / Failed	Skipped (connection)

Failures (19)

Failure Cluster	Category	Failures
Tool Execution Failures	Tool Execution	9
LLM Reasoning Issues	Llm Reasoning	6
Data Availability Issues	Data Availability	4

Tool Execution Failures TOOL EXECUTION (9 failures)

1. Noor_Al_aqaba

Production • 78.0s

Question: Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

Error reason: AI indicated failure (matched 'unexpected keyword argument'): Clipping Loss Calculation: - Data Quality Note: The clipping-loss tool failed internally and couldn't calculate inverter-by-inverter clipping losses for the last 6 months. - Error encountered: calcul

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "Please figure out how much clipping loss was found in the last 6 months broken by invert
ers, list them down and plot it as well",
    "displayed_response": "Clipping Loss Calculation:\n\n- Data Quality Note: The clipping-loss tool failed
internally and couldn’t calculate inverter-by-inverter clipping losses for the last 6 months.\n- Error enco
untered: calculate_clipping_or_curtailment_losses() got an unexpected keyword argument 'reactive_col'\n\nVi
sualization:\n\n- The graph “Clipping Loss by Inverter – Last 6 Months” has been prepared.",
    "coding_response": [
      "I couldn’t calculate the clipping loss because the clipping-loss tool failed internally with an exce
ption.\n\nThe tool error was:\n`calculate_clipping_or_curtailment_losses() got an unexpected keyword argume
nt 'reactive_col'`\n\nSo I can’t provide a trustworthy inverter-by-inverter clipping-loss breakdown from th
e available tooling right now.",
      {
        "type": "graph",
        "series": [
          "",
          "",
          "",
          ""
        ],
        "graphTitle": "Clipping Loss by Inverter - Last 6 Months",
        "yAxisTitle": "Clipping Loss (kWh)"
      }
    ],
    "used_agents": [
      "data_agent",
      "graph_agent"
    ],
    "used_tools": [
      "total_energy_func_ag",
      "calculate_clipping_or_curtailment_losses_ag",
      "functions.inspect_xpool",
      "functions.calculate_clipping_or_curtailment_losses_ag"
    ],
    "urls": [],
    "steps": "1. I filter the data to include only records from February 5, 2026, to August 5, 2026 (includ
ing the end date).\n2. I find all relevant inverter power columns in the dataset, making sure to leave out
any columns related to reactive power or status flags.\n3. I check that the required power columns are actu
ally present in the data.\n4. I clean up the plane of array (POA) irradiance data by filling in missing val
ues using a method that accounts for daily sunlight patterns, to keep the data reliable.\n5. For each inver
ter power column, I calculate energy losses by finding the absolute difference between the measured power a
nd the inverter’s rated AC power during periods when losses are detected.\n   - I identify clipping events
using both a machine learning model (rdtools XGBoost clip filter) and a simple rule: if the power is above
85% of the inverter’s rated AC power, it’s considered clipped.\n6. I convert these power losses into energy
lost (in kWh) by multiplying the loss by the time interval (0.0833 hours).\n7. I group the energy loss valu
es by the chosen time period (such as day, month, or year) for each inverter.\n8. I provide a detailed brea
kdown of the losses for each inverter and each time period.\n9. I organize the results into a table, with i
nverters as columns and time periods as rows.\n10. I return the results as a dictionary, either broken down
by time period or as a single total, depending on the settings."
  }
}
```

AEPCO Alternative Energy Projects Company

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Clipping Loss Calculation:

- Data Quality Note: The clipping-loss tool failed internally and couldn't calculate inverter-by-inverter clipping losses for the last 6 months.
- Error encountered: calculate_clipping_or_curtailment_losses() got an unexpected keyword argument 'reactive_col'

Visualization:

- The graph "Clipping Loss by Inverter – Last 6 Months" has been prepared.

08:05 AM

Are you satisfied with the answer?

Time taken: 63s, token: 0 tokens

connected to Noor Al Aqaba

How can we help you today?

LEAVE ANY FEEDBACK

Mistakes in P2Chat are possible. Double-check key information.

2. Noor_Al_aqaba

Production • 64.8s

Question: Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

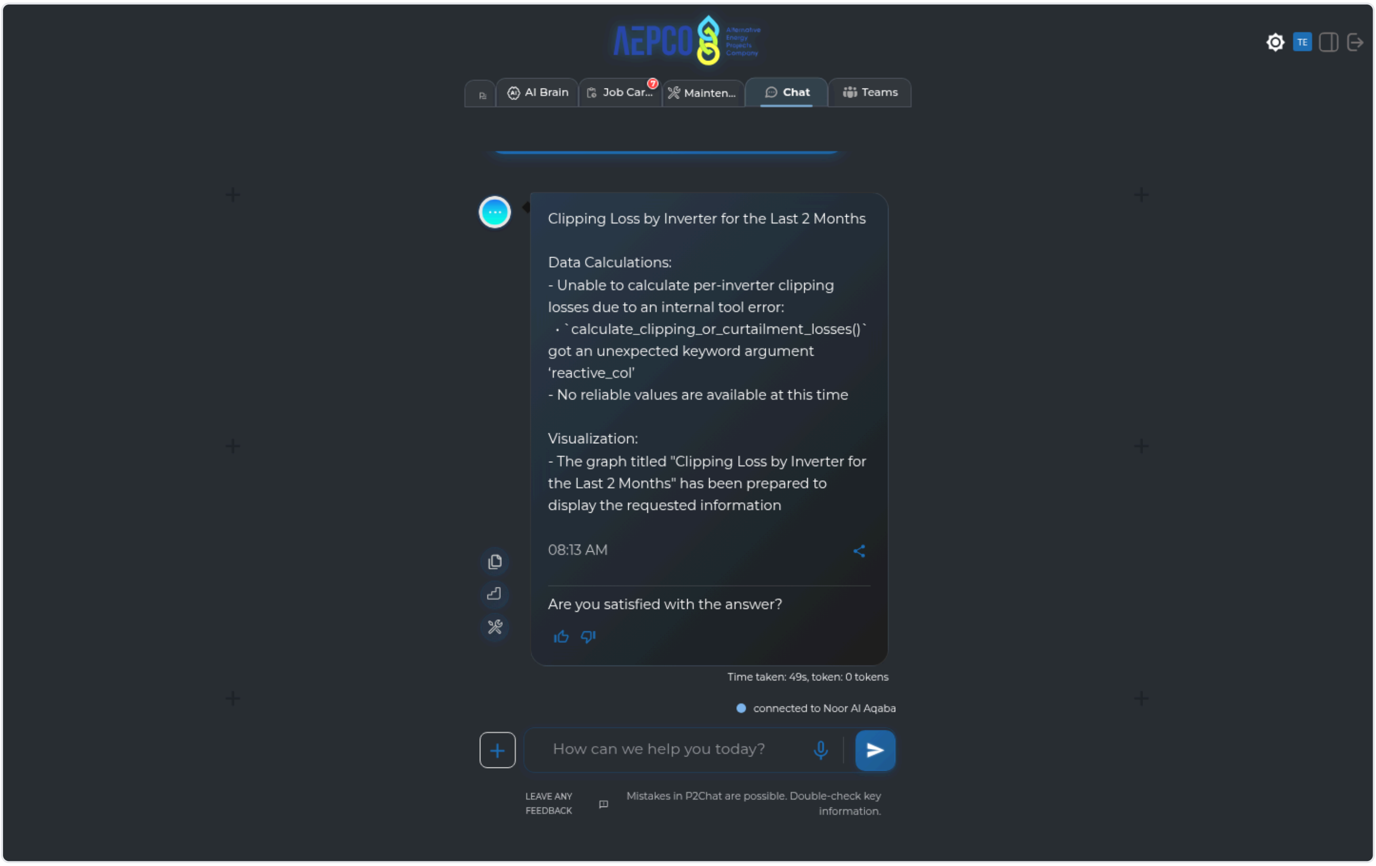
Error reason: AI indicated failure (matched 'tool error'): Clipping Loss by Inverter for the Last 2 Months Data Calculations: - Unable to calculate per-inverter clipping losses due to an internal tool error: • `calculate_clipping_or_curtailment_losses()` g

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well",
    "displayed_response": "Clipping Loss by Inverter for the Last 2 Months\n\nData Calculations:\n- Unable to calculate per-inverter clipping losses due to an internal tool error:\n  • `calculate_clipping_or_curtailment_losses()` got an unexpected keyword argument ‘reactive_col’\n- No reliable values are available at this time\n\nVisualization:\n- The graph titled \"Clipping Loss by Inverter for the Last 2 Months\" has been prepared to display the requested information",
    "coding_response": [
      "I couldn’t calculate the clipping losses because the clipping-loss tool failed internally with an exception.\n\nError returned:\n`calculate_clipping_or_curtailment_losses() got an unexpected keyword argument 'reactive_col'`\n\nSo I’m unable to provide reliable per-inverter clipping-loss values for the last 2 months from the available tools right now.",
      {
        "type": "graph",
        "series": [
          "",
          "",
          "",
          ""
        ],
        "graphTitle": "Clipping Loss by Inverter for the Last 2 Months",
        "yAxisTitle": "Clipping Loss (kWh)"
      }
    ],
    "used_agents": [
      "data_agent",
      "graph_agent"
    ],
    "used_tools": [
      "functions.inspect_xpool",
      "functions.calculate_clipping_or_curtailment_losses_ag",
      "functions.total_energy_func_ag"
    ],
    "urls": [],
    "steps": "1. I filter the data to include only records from June 5, 2026, to August 5, 2026, including the last day.\n\n2. I find all inverter power columns in the dataset, making sure to leave out any columns related to reactive power or flags.\n\n3. I check that the required power columns are actually present in the data.\n\n4. I clean up the plane of array (POA) irradiance data by filling in any missing values using a method that accounts for daily sunlight patterns.\n\n5. For each inverter, I calculate power losses by finding the absolute difference between the measured power and the inverter’s rated AC power during periods when losses are detected.\n   - I identify clipping events using both a machine learning approach (rdtools XGBoost clip filter) and a simple rule: if power is above 85% of the rated AC power, it’s considered clipped.\n\n6. I convert these power losses into energy losses (in kWh) by multiplying by the time interval (0.0833 hours).\n\n7. I group the energy loss values by the chosen time period (such as day, month, or year) for each inverter.\n\n8. I provide a detailed breakdown of the loss values for each inverter and each time period.\n\n9. I organize the results into a DataFrame, with each inverter as a column and each time period as a row.\n\n10. I return the results as a dictionary, where each key is a time period (or a single total, depending on the settings)."
  }
}
```


AI Brain
Job Car...
Mainten...
Chat
Teams
Clipping Loss by Inverter for the Last 2 Months
Data Calculations:
- Unable to calculate per-inverter clipping losses due to an internal tool error:
• `calculate_clipping_or_curtailment_losses()` got an unexpected keyword argument 'reactive_col'
- No reliable values are available at this time
Visualization:
- The graph titled "Clipping Loss by Inverter for the Last 2 Months" has been prepared to display the requested information
08:13 AM
Are you satisfied with the answer?
Time taken: 49s, token: 0 tokens
connected to Noor Al Aqaba
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

3. Galleria_Mall

Production • 99.3s

Question: Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

Error reason: AI indicated failure (matched 'unexpected keyword argument'): Clipping Loss Calculation Summary (Last 6 Months: 2026-02-05 to 2026-08-05) Data Retrieval & Quality Notes: - Attempted calculation of per-inverter clipping losses using the clipping-loss tool.

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "Please figure out how much clipping loss was found in the last 6 months broken by invert
ers, list them down and plot it as well",
    "displayed_response": "Clipping Loss Calculation Summary (Last 6 Months: 2026-02-05 to 2026-08-05)\n\nD
ata Retrieval & Quality Notes:\n- Attempted calculation of per-inverter clipping losses using the clipping-
loss tool.\n- Tool failed internally with error: `unexpected keyword argument 'reactive_col'`.\n- No per-in
verter clipping totals could be produced for the specified period.\n\nVisualization:\n- The visualization h
as been prepared (Clipping Loss by Inverter Over the Last 6 Months (Monthly)).\n",
    "coding_response": [
      "I couldn’t calculate clipping losses because the clipping-loss tool is currently failing internally
for this site: it returns an error (`unexpected keyword argument 'reactive_col'`) instead of values.\n\nI a
ttempted the last 6 months window (2026-02-05 to 2026-08-05), but no per-inverter clipping totals were prod
uced.\n\nCalled tools: calculate_clipping_or_curtailment_losses_ag twice with clipping settings; both faile
d with the same internal error.",
      {
        "type": "graph",
        "series": [
          "<html>\n<head><meta charset=\"utf-8\" /></head>\n<body>\n    <div>                        <scrip
t type=\"text/javascript\">window.PlotlyConfig = {MathJaxConfig: 'local'};</script>\n        <script charse
t=\"utf-8\" src=\"https://cdn.plot.ly/plotly-3.0.1.min.js\"></script>                <div id=\"613f358b-d4a
2-4904-afca-29f4d6c37443\" class=\"plotly-graph-div\" style=\"height:100%; width:100%;\"></div>
<script type=\"text/javascript\">                window.PLOTLYENV=window.PLOTLYENV || {};
if (document.getElementById(\"613f358b-d4a2-4904-afca-29f4d6c37443\")) {                    Plotly.newPlot(
\"613f358b-d4a2-4904-afca-29f4d6c37443\",                        [{\"hovertemplate\":\"%{fullData.name}: %
{y:.2f} kWh\\u003cextra\\u003e\\u003c\\u002fextra\\u003e\",\"name\":\"Inverter 10\",\"x\":[\"2026-04-01T00:
00:00\",\"2026-05-01T00:00:00\",\"2026-06-01T00:00:00\",\"2026-07-01T00:00:00\"],\"y\":{\"dtype\":\"f8
\",\"bdata\":\"G3rYF6JJEUBPTTsE3vQhQBcnLjEn07g\\u002fAAAAAAAAAAA=\"},\"type\":\"bar\"},{\"hovertemplate
\":\"%{fullData.name}: %{y:.2f} kWh\\u003cextra\\u003e\\u003c\\u002fextra\\u003e\",\"name\":\"Inverter 11
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\"],\"y\":{\"dtype\":\"f8\",\"bdata\":\"8Gvt9+tpJUDXn7BI2schQKZj\\u002fZDO7uA\\u002f\\u002fJIbU00P6D8=
\"},\"type\":\"bar\"},{\"hovertemplate\":\"%{fullData.name}: %{y:.2f} kWh\\u003cextra\\u003e\\u003c\\u002fe
xtra\\u003e\",\"name\":\"Inverter 12\",\"x\":[\"2026-04-01T00:00:00\",\"2026-05-01T00:00:00\",\"2026-06-01T
00:00:00\",\"2026-07-01T00:00:00\"],\"y\":{\"dtype\":\"f8\",\"bdata\":\"WPSj3+1VDEBS7m\\u002fZVVcdQDplBMUga
9I\\u002fAAAAAAAAAAA=\"},\"type\":\"bar\"},{\"hovertemplate\":\"%{fullData.name}: %{y:.2f} kWh\\u003cextra
\\u003e\\u003c\\u002fextra\\u003e\",\"name\":\"Inverter 1\",\"x\":[\"2026-04-01T00:00:00\",\"2026-05-01T00:
00:00\",\"2026-06-01T00:00:00\",\"2026-07-01T00:00:00\"],\"y\":{\"dtype\":\"f8\",\"bdata\":\"grnGxdgQJEBYKL
cDj7sjQAAAAAAAAAAAAAAAAAAAAAA=\"},\"type\":\"bar\"},{\"hovertemplate\":\"%{fullData.name}: %{y:.2f} kWh\\u0
03cextra\\u003e\\u003c\\u002fextra\\u003e\",\"name\":\"Inverter 2\",\"x\":[\"2026-04-01T00:00:00\",\"2026-0
5-01T00:00:00\",\"2026-06-01T00:00:00\",\"2026-07-01T00:00:00\"],\"y\":{\"dtype\":\"f8\",\"bdata\":\"vX77zZ
bXCkCBdL8BCkEXQCSh91jiZ9I\\u002fAAAAAAAAAAA=\"},\"type\":\"bar\"},{\"hovertemplate\":\"%{fullData.name}: %
{y:.2f} kWh\\u003cextra\\u003e\\u003c\\u002fextra\\u003e\",\"name\":\"Inverter 3\",\"x\":[\"2026-04-01T00:0
0:00\",\"2026-05-01T00:00:00\",\"2026-06-01T00:00:00\",\"2026-07-01T00:00:00\"],\"y\":{\"dtype\":\"f8\",\"b
data\":\"xpLDEm6h8T8WSzXLc3MdQFpzmoILQd4\\u002fAAAAAAAAAAA=\"},\"type\":\"bar\"},{\"hovertempl
```
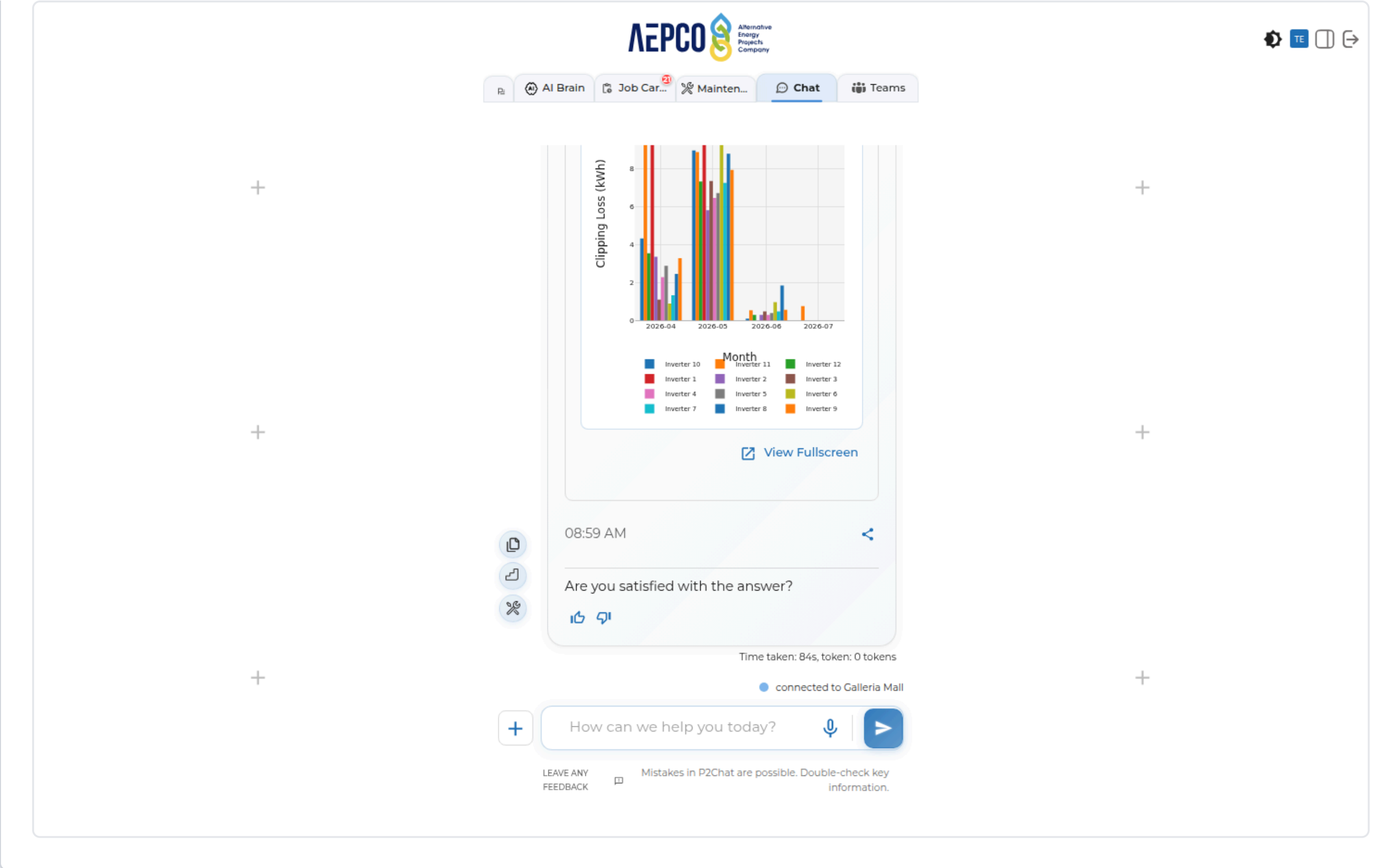


Console errors: An iframe which has both allow-scripts and allow-same-origin for its sandbox attribute can escape its sandboxing.



AEPCO
Alternative Energy Projects Company
AI Brain
Job Car...
Mainten...
Chat
Teams
Clipping Loss (kWh)
2026-04
2026-05
2026-06
2026-07
Month
Inverter 10
Inverter 11
Inverter 12
Inverter 1
Inverter 2
Inverter 3
Inverter 4
Inverter 5
Inverter 6
Inverter 7
Inverter 8
Inverter 9
View Fullscreen
08:59 AM
Are you satisfied with the answer?
Time taken: 84s, token: 0 tokens
connected to Galleria Mall
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

4. Galleria_Mall

Production • 60.0s

Question: Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

Error reason: AI indicated failure (matched 'unexpected keyword argument'): Clipping Loss Breakdown (Last 2 Months): - Unable to provide inverter-by-inverter clipping losses due to an internal calculation error (calculate_clipping_or_curtailment_losses_ag failed with `unexp

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{"result":"{\n    \"user_input\": \"Please figure out how much clipping loss was found in the last 2 months
broken by inverters, list them down and plot it as well\",\n    \"displayed_response\": \"Clipping Loss Bre
akdown (Last 2 Months):\\n- Unable to provide inverter-by-inverter clipping losses due to an internal calcu
lation error (calculate_clipping_or_curtailment_losses_ag failed with `unexpected keyword argument 'reactiv
e_col'`).\\n\\nVisualization:\\n- The visualization has been prepared.\",\n    \"coding_response\": [\n
\"I couldn\\u2019t calculate clipping loss because the clipping-loss tool failed internally with an excepti
on.\\n\\nError returned:\\n- `calculate_clipping_or_curtailment_losses_ag` failed: `unexpected keyword argu
ment 'reactive_col'`\\n\\nSo I can\\u2019t provide a trustworthy inverter-by-inverter clipping loss breakdo
wn for the last 2 months from the available tools right now.\",\n        {\n            \"type\": \"graph
\",\n            \"series\": [\n                \"<html>\\n<head><meta charset=\\\"utf-8\\\" /></head>\\n<b
ody>\\n    <div>                        <script type=\\\"text/javascript\\\">window.PlotlyConfig = {MathJax
Config: 'local'};</script>\\n        <script charset=\\\"utf-8\\\" src=\\\"https://cdn.plot.ly/plotly-3.0.
1.min.js\\\"></script>                <div id=\\\"b935a759-b038-4bfc-a1bf-ac2a81d71d71\\\" class=\\\"plotly
-graph-div\\\" style=\\\"height:100%; width:100%;\\\"></div>            <script type=\\\"text/javascript
\\\">                window.PLOTLYENV=window.PLOTLYENV || {};                                    if (document.g
etElementById(\\\"b935a759-b038-4bfc-a1bf-ac2a81d71d71\\\")) {                    Plotly.newPlot(
\\\"b935a759-b038-4bfc-a1bf-ac2a81d71d71\\\",                        [{\\\"hovertemplate\\\":\\\"%{fullDat
a.name}: %{y:.2f} kWh\\\\u003cextra\\\\u003e\\\\u003c\\\\u002fextra\\\\u003e\\\",\\\"mode\\\":\\\"lines+mar
kers\\\",\\\"name\\\":\\\"Inverter 10\\\",\\\"x\\\":[\\\"2026-06-05T00:00:00\\\",\\\"2026-06-06T00:00:00
\\\",\\\"2026-06-07T00:00:00\\\",\\\"2026-06-08T00:00:00\\\",\\\"2026-06-09T00:00:00\\\",\\\"2026-06-10T00:
00:00\\\",\\\"2026-06-11T00:00:00\\\",\\\"2026-06-12T00:00:00\\\",\\\"2026-06-13T00:00:00\\\",\\\"2026-06-1
4T00:00:00\\\",\\\"2026-06-15T00:00:00\\\",\\\"2026-06-16T00:00:00\\\",\\\"2026-06-17T00:00:00\\\",\\\"2026
-06-18T00:00:00\\\",\\\"2026-06-19T00:00:00\\\",\\\"2026-06-20T00:00:00\\\",\\\"2026-06-21T00:00:00
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00:00\\\",\\\"2026-06-26T00:00:00\\\",\\\"2026-06-27T00:00:00\\\",\\\"2026-06-28T00:00:00\\\",\\\"2026-06-2
9T00:00:00\\\",\\\"2026-06-30T00:00:00\\\",\\\"2026-07-01T00:00:00\\\",\\\"2026-07-02T00:00:00\\\",\\\"2026
-07-03T00:00:00\\\",\\\"2026-07-04T00:00:00\\\",\\\"2026-07-05T00:00:00\\\",\\\"2026-07-06T00:00:00
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00:00\\\",\\\"2026-07-11T00:00:00\\\",\\\"2026-07-12T00:00:00\\\",\\\"2026-07-13T00:00:00\\\",\\\"2026-07-1
4T00:00:00\\\",\\\"2026-07-15T00:00:00\\\",\\\"2026-07-16T00:00:00\\\",\\\"2026-07-17T00:00:00\\\",\\\"2026
-07-18T00:00:00\\\",\\\"2026-07-19T00:00:00\\\",\\\"2026-07-20T00:00:00\\\",\\\"2026-07-21T00:00:00
\\\",\\\"2026-07-22T00:00:00\\\",\\\"2026-07-23T00:00:00\\\",\\\"2026-07-24T00:00:00\\\",\\\"2026-07-25T00:
00:00\\\",\\\"2026-07-26T00:00:00\\\",\\\"2026-07-27T00:00:00\\\",\\\"2026-07-28T00:00:00\\\",\\\"2026-07-2
9T00:00:00\\\"],\\\"y\\\":{\\\"dtype\\\":\\\"f8\\\",\\\"bdata\\\":\\\"AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
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AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAADx0k1iELSJQJLtfD8
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dAvQLkXnPKd0CPbOf7qSd4QMd2vp+aUnhA9OEz7IrOdkDBPFH\\\\u00
```
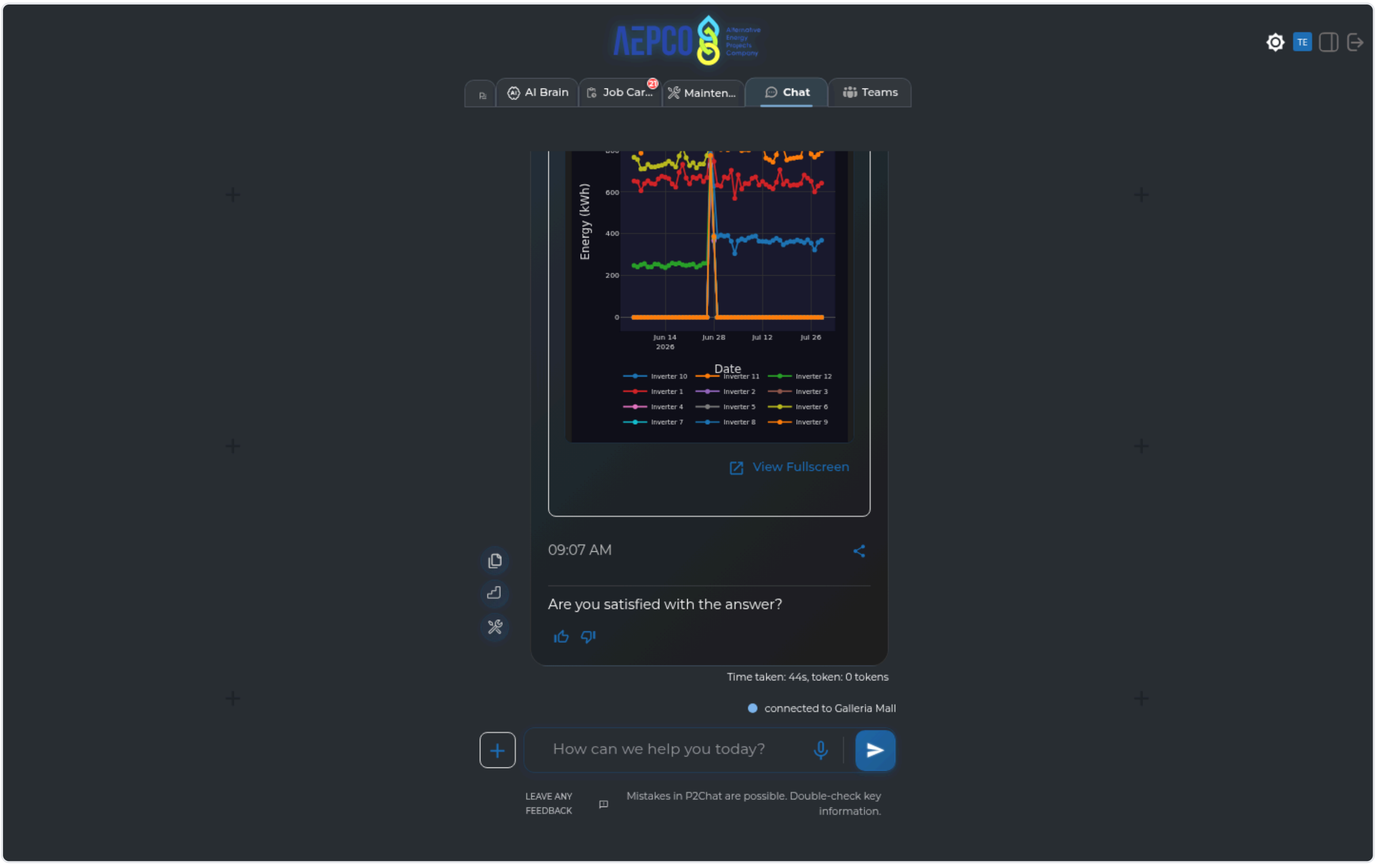


Console errors: An iframe which has both allow-scripts and allow-same-origin for its sandbox attribute can escape its sandboxing.


AEPCO
AI Brain
Job Car...
Mainten...
Chat
Teams
Energy (kWh)
600
400
200
0
Jun 14
2026
Jun 28
Jul 12
Jul 26
Date
Inverter 10
Inverter 11
Inverter 12
Inverter 1
Inverter 2
Inverter 3
Inverter 4
Inverter 5
Inverter 6
Inverter 7
Inverter 8
Inverter 9
View Fullscreen
09:07 AM
Are you satisfied with the answer?
Time taken: 44s, token: 0 tokens
connected to Galleria Mall
How can we help you today?
LEAVE ANY
FEEDBACK
Mistakes in P2Chat are possible. Double-check key
information.

5. GID

Production • 75.5s

Question: Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

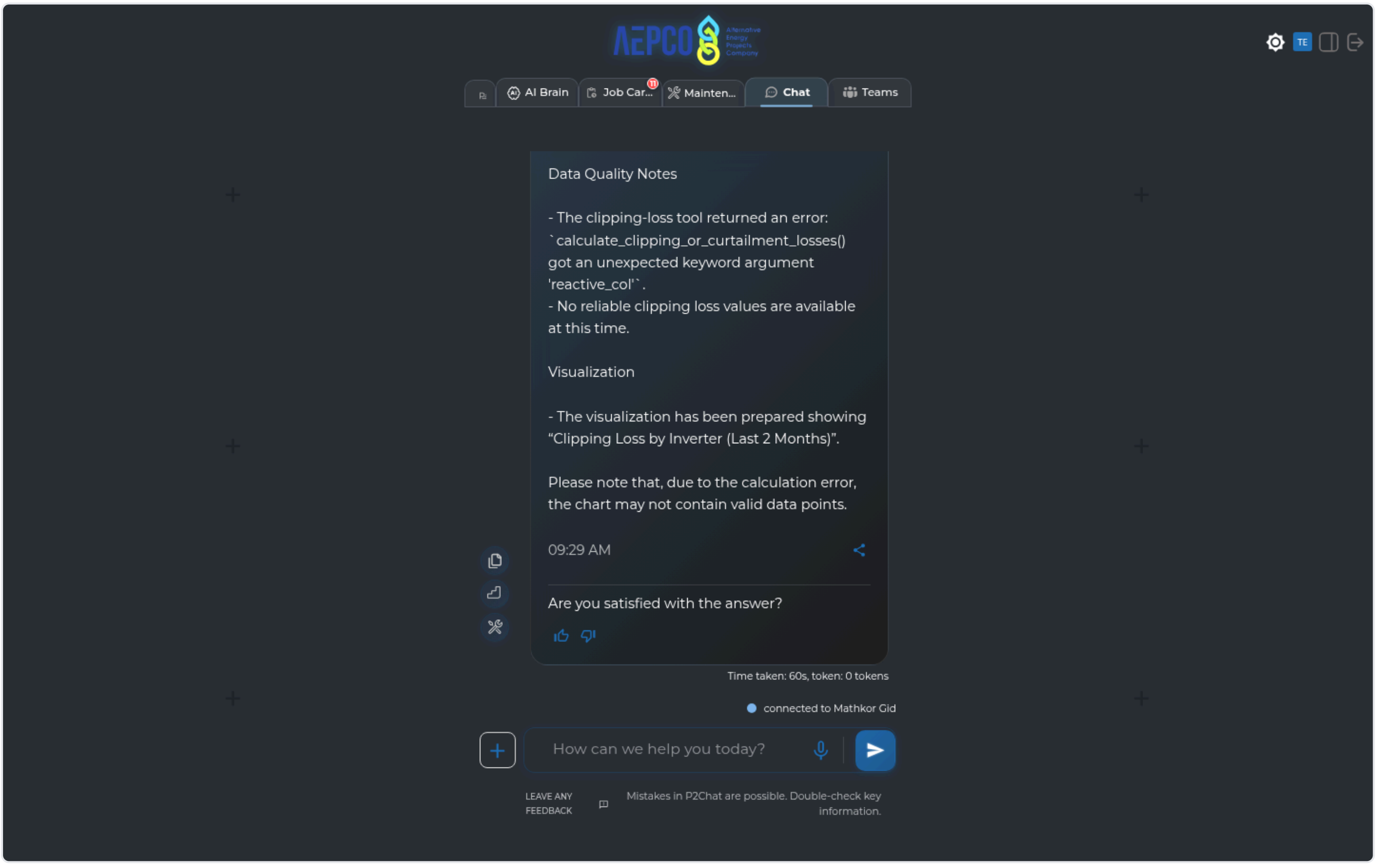
Error reason: AI indicated failure (matched 'an error'): Data Summary - We attempted to calculate inverter-by-inverter clipping loss for the last 2 months, but the calculation tool failed internally. Data Quality Notes - The clipping-loss tool returned a

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "Please figure out how much clipping loss was found in the last 2 months broken by invert
ers, list them down and plot it as well",
    "displayed_response": "Data Summary\n\n- We attempted to calculate inverter-by-inverter clipping loss f
or the last 2 months, but the calculation tool failed internally.\n\nData Quality Notes\n\n- The clipping-l
oss tool returned an error: `calculate_clipping_or_curtailment_losses() got an unexpected keyword argument
'reactive_col'`.\n- No reliable clipping loss values are available at this time.\n\nVisualization\n\n- The
visualization has been prepared showing “Clipping Loss by Inverter (Last 2 Months)”.\n\nPlease note that, d
ue to the calculation error, the chart may not contain valid data points.",
    "coding_response": [
      "I couldn’t calculate the clipping loss because the clipping-loss tool failed internally for this req
uest.\n\nThe tool returned this error:\n`calculate_clipping_or_curtailment_losses() got an unexpected keywo
rd argument 'reactive_col'`\n\nSo I’m unable to provide a reliable inverter-by-inverter clipping loss break
down for the last 2 months from the available tooling right now.",
      {
        "type": "graph",
        "series": [
          "",
          "",
          "",
          ""
        ],
        "graphTitle": "Clipping Loss by Inverter (Last 2 Months)",
        "yAxisTitle": "Clipping Loss (kWh)"
      }
    ],
    "used_agents": [
      "data_agent",
      "graph_agent"
    ],
    "used_tools": [
      "functions.calculate_clipping_or_curtailment_losses_ag",
      "functions.inspect_xpool",
      "total_energy_func_ag",
      "calculate_clipping_or_curtailment_losses_ag"
    ],
    "urls": [],
    "steps": "1. I filter the data to include only records from June 5, 2026, to August 5, 2026, including
the last day.\n2. I find all relevant inverter power columns, making sure to leave out any columns related
to reactive power or status flags.\n3. I check that the chosen power columns are actually present in the da
taset.\n4. I clean up the plane of array (POA) irradiance data by filling in missing values using a method
that accounts for daily sunlight patterns.\n5. For each inverter power column, I calculate energy losses by
finding the absolute difference between the measured power and the inverter’s rated AC power during periods
of detected losses.\n   - I identify clipping events using both a machine learning model (rdtools XGBoost c
lip filter) and a rule that flags power readings above 85% of the inverter’s rated AC power.\n6. I convert
these power losses into energy lost (in kWh) by multiplying by the time interval of 0.0833 hours.\n7. I sum
up the energy losses for each inverter, grouping them by the chosen time period (such as day, month, or yea
r).\n8. I provide a detailed breakdown of loss values for each inverter and each time period.\n9. I organiz
e the results into a DataFrame, with inverters as columns and time periods as rows.\n10. I return the resul
ts as a dictionary, either with each time period as a key or as a single total, depending on the setting
s.\n\nFor total energy calculation:\n1. I select data from June 5, 2026, to August 5, 2026, including the e
nd date.\n2. I identify all inverter power columns, excluding those for reactive power or flags.\n3. I clea
n the data by removing or flagging any invalid measurements in each inverter’s power column.\n4. For each i
nverter, I calculate energy by multiplying the power by the time interval (0.0833 hours) and adjusting for
units if needed (e.g., dividing by 1000 if the data is in Watts).\n5. I include all days in the calculation
for each inverter.\n6. I sum up the energy values for each inverter, grouping them by the chosen time perio
d (such as day, month, or year).\n7. I provide the energy value for each inverter and each time period.\n8.
I organize the results into a DataFrame, with inverters as columns and time periods as rows.\n9. I return t
he results as a dictionary,
```


AEPCO
AI Brain
Job Car...
Mainten...
Chat
Teams
Data Quality Notes
- The clipping-loss tool returned an error: `calculate_clipping_or_curtailment_losses() got an unexpected keyword argument 'reactive_col'`.
- No reliable clipping loss values are available at this time.
Visualization
- The visualization has been prepared showing "Clipping Loss by Inverter (Last 2 Months)".
Please note that, due to the calculation error, the chart may not contain valid data points.
09:29 AM
Are you satisfied with the answer?
Time taken: 60s, token: 0 tokens
connected to Mathkor Gid
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

6. BF_Wheeling

Production • 44.4s

Question: Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

Error reason: AI indicated failure (matched 'tool error'): Clipping Loss Calculation (Last 6 Months: 2026-02-05 to 2026-08-05) - Inverters analyzed: inverter1 through inverter6 - Status: Calculation could not be completed due to an internal tool error • Er

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "Please figure out how much clipping loss was found in the last 6 months broken by invert
ers, list them down and plot it as well",
    "displayed_response": "Clipping Loss Calculation (Last 6 Months: 2026-02-05 to 2026-08-05)\n\n- Inverte
rs analyzed: inverter1 through inverter6\n- Status: Calculation could not be completed due to an internal t
ool error\n  • Error details: `calculate_clipping_or_curtailment_losses()` got an unexpected keyword argume
nt `reactive_col`\n- Data quality note: No reliable clipping-loss values are available at this time\n\nVisu
alization\n\n- The chart showing clipping loss by inverter over the last 6 months has been prepared.",
    "coding_response": [
      "I couldn’t calculate the clipping losses because the clipping-loss tool failed internally with an ex
ception.\n\nThe failed request was for the last 6 months (2026-02-05 to 2026-08-05) and per inverter.\n\nEr
ror returned:\n- `calculate_clipping_or_curtailment_losses() got an unexpected keyword argument 'reactive_c
ol'`\n\nSo I’m unable to provide trustworthy clipping-loss values for inverter1 through inverter6 from the
available tools right now.",
      {
        "type": "graph",
        "series": [
          "",
          "",
          "",
          ""
        ],
        "graphTitle": "Clipping loss by inverter over the last 6 months",
        "yAxisTitle": "Loss (kWh)"
      }
    ],
    "used_agents": [
      "data_agent",
      "graph_agent"
    ],
    "used_tools": [
      "total_energy_func_ag",
      "functions.inspect_xpool",
      "functions.calculate_clipping_or_curtailment_losses_ag",
      "calculate_clipping_or_curtailment_losses_ag"
    ],
    "urls": [],
    "steps": "1. I filter the data to include only records from February 5, 2026, to August 5, 2026, includ
ing the end date.\n\n2. I find all relevant inverter power columns in the dataset, making sure to leave out
any columns related to reactive power or status flags.\n\n3. I check that the power columns I want to use a
re actually present in the dataset.\n\n4. I clean up the plane of array (POA) irradiance data by filling in
any missing values using a method that accounts for daily sunlight patterns, to keep the data accurate.\n\n
5. For each inverter power column:\n   - I calculate energy losses by taking the absolute difference betwee
n the measured power and the inverter’s rated AC power during times when losses are detected.\n   - I ident
ify clipping events using both a machine learning model (rdtools XGBoost clip filter) and a simple rule: if
the power is above 85% of the rated AC power, it’s considered clipped.\n\n6. I convert the detected power l
osses into energy lost (in kWh) by multiplying by the time interval (0.0833 hours).\n\n7. I group the energ
y loss values by the chosen time period (such as day, month, or year) for each inverter.\n\n8. I add up the
grouped values to get a total loss for each inverter over the selected period.\n\n9. I organize the results
into a DataFrame, with each inverter as a column and each time period as a row.\n\n10. I return the results
as a dictionary, either broken down by time period or as a single total value per inverter, depending on th
e settings."
  }
}
```

AI Brain | Job Car... | Mainten... | Chat | Teams

02-05 to 2026-08-05)

- Inverters analyzed: inverter1 through inverter6
- Status: Calculation could not be completed due to an internal tool error
  - Error details: `calculate_clipping_or_curtailment_losses()` got an unexpected keyword argument `reactive_col`
- Data quality note: No reliable clipping-loss values are available at this time

Visualization

- The chart showing clipping loss by inverter over the last 6 months has been prepared.

09:56 AM

Are you satisfied with the answer?

Time taken: 29s, token: 0 tokens

connected to Bf Wheeling

How can we help you today?

LEAVE ANY FEEDBACK

Mistakes in P2Chat are possible. Double-check key information.

7. BF_Wheeling

Production • 42.2s

Question: Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

Error reason: AI indicated failure (matched 'unexpected keyword argument'): - Clipping-loss calculation for the period 2026-06-05 to 2026-08-05 failed; no inverter-level clipping losses could be returned. • Error raised by the tool: `calculate_clipping_or_curtailment_losse

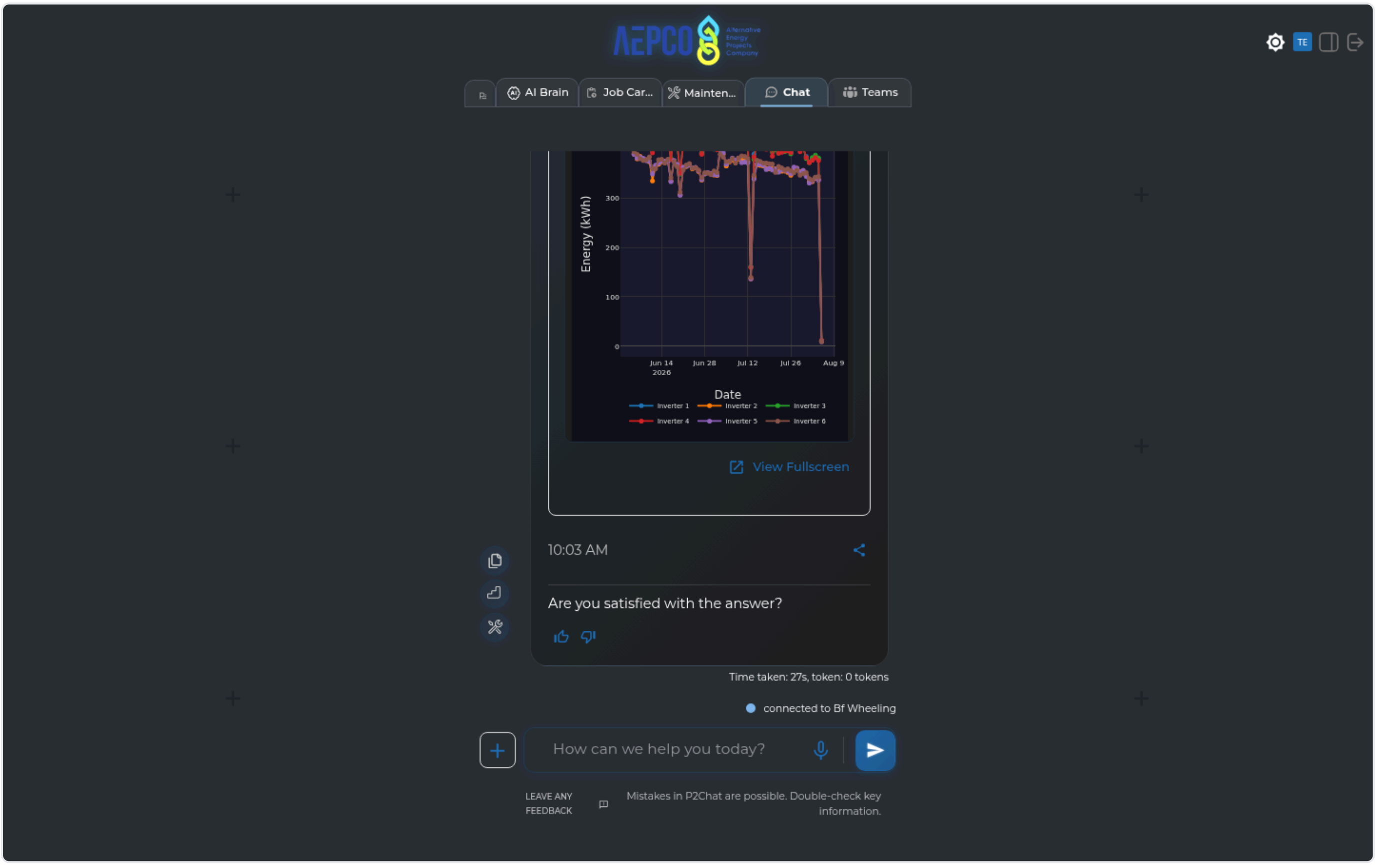
TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{"result":"{\n    \"user_input\": \"Please figure out how much clipping loss was found in the last 2 months
broken by inverters, list them down and plot it as well\",\n    \"displayed_response\": \"- Clipping-loss c
alculation for the period 2026-06-05 to 2026-08-05 failed; no inverter\\u2010level clipping losses could be
returned.\\n\\n\\u2022 Error raised by the tool: `calculate_clipping_or_curtailment_losses() got an unexpec
ted keyword argument 'reactive_col'`\\n\\n- As a fallback, a daily energy production chart by inverter has
been prepared for that same window.\\n\\nI\\u2019ve created a line chart showing daily inverter energy for
the requested period.\",\n    \"coding_response\": [\n        \"I couldn\\u2019t calculate clipping loss be
cause the clipping-loss tool is currently failing internally for this site/query. It errors before returnin
g inverter values.\\n\\nFailed query window: 2026-06-05 to 2026-08-05\\nRequested breakdown: inverter1, inv
erter2, inverter3, inverter4, inverter5, inverter6\\n\\nThe tool raised: `calculate_clipping_or_curtailment
_losses() got an unexpected keyword argument 'reactive_col'`\",\n        {\n            \"type\": \"graph
\",\n            \"series\": [\n                \"<html>\\n<head><meta charset=\\\"utf-8\\\" /></head>\\n<b
ody>\\n    <div>                        <script type=\\\"text/javascript\\\">window.PlotlyConfig = {MathJax
Config: 'local'};</script>\\n        <script charset=\\\"utf-8\\\" src=\\\"https://cdn.plot.ly/plotly-3.0.
1.min.js\\\"></script>                <div id=\\\"18bab5d8-1611-4cc8-b29d-f46c60e4eb1b\\\" class=\\\"plotly
-graph-div\\\" style=\\\"height:100%; width:100%;\\\"></div>            <script type=\\\"text/javascript
\\\">                window.PLOTLYENV=window.PLOTLYENV || {};                                if (document.g
etElementById(\\\"18bab5d8-1611-4cc8-b29d-f46c60e4eb1b\\\")) {                    Plotly.newPlot(
\\\"18bab5d8-1611-4cc8-b29d-f46c60e4eb1b\\\",                        [{\\\"hovertemplate\\\":\\\"%{fullDat
a.name}: %{y:.2f} kWh\\\\u003cextra\\\\u003e\\\\u003c\\\\u002fextra\\\\u003e\\\",\\\"mode\\\":\\\"lines+mar
kers\\\",\\\"name\\\":\\\"Inverter 1\\\",\\\"x\\\":[\\\"2026-06-05T00:00:00\\\",\\\"2026-06-06T00:00:00
\\\",\\\"2026-06-07T00:00:00\\\",\\\"2026-06-08T00:00:00\\\",\\\"2026-06-09T00:00:00\\\",\\\"2026-06-10T00:
00:00\\\",\\\"2026-06-11T00:00:00\\\",\\\"2026-06-12T00:00:00\\\",\\\"2026-06-13T00:00:00\\\",\\\"2026-06-1
4T00:00:00\\\",\\\"2026-06-15T00:00:00\\\",\\\"2026-06-16T00:00:00\\\",\\\"2026-06-17T00:00:00\\\",\\\"2026
-06-18T00:00:00\\\",\\\"2026-06-19T00:00:00\\\",\\\"2026-06-20T00:00:00\\\",\\\"2026-06-21T00:00:00
\\\",\\\"2026-06-22T00:00:00\\\",\\\"2026-06-23T00:00:00\\\",\\\"2026-06-24T00:00:00\\\",\\\"2026-06-25T00:
00:00\\\",\\\"2026-06-26T00:00:00\\\",\\\"2026-06-27T00:00:00\\\",\\\"2026-06-28T00:00:00\\\",\\\"2026-06-2
9T00:00:00\\\",\\\"2026-06-30T00:00:00\\\",\\\"2026-07-01T00:00:00\\\",\\\"2026-07-02T00:00:00\\\",\\\"2026
-07-03T00:00:00\\\",\\\"2026-07-04T00:00:00\\\",\\\"2026-07-05T00:00:00\\\",\\\"2026-07-06T00:00:00
\\\",\\\"2026-07-07T00:00:00\\\",\\\"2026-07-08T00:00:00\\\",\\\"2026-07-09T00:00:00\\\",\\\"2026-07-10T00:
00:00\\\",\\\"2026-07-11T00:00:00\\\",\\\"2026-07-12T00:00:00\\\",\\\"2026-07-13T00:00:00\\\",\\\"2026-07-1
4T00:00:00\\\",\\\"2026-07-15T00:00:00\\\",\\\"2026-07-16T00:00:00\\\",\\\"2026-07-17T00:00:00\\\",\\\"2026
-07-18T00:00:00\\\",\\\"2026-07-19T00:00:00\\\",\\\"2026-07-20T00:00:00\\\",\\\"2026-07-21T00:00:00
\\\",\\\"2026-07-22T00:00:00\\\",\\\"2026-07-23T00:00:00\\\",\\\"2026-07-24T00:00:00\\\",\\\"2026-07-25T00:
00:00\\\",\\\"2026-07-26T00:00:00\\\",\\\"2026-07-27T00:00:00\\\",\\\"2026-07-28T00:00:00\\\",\\\"2026-07-2
9T00:00:00\\\",\\\"2026-07-30T00:00:00\\\",\\\"2026-07-31T00:00:00\\\",\\\"2026-08-01T00:00:00\\\",\\\"2026
-08-02T00:00:00\\\",\\\"2026-08-03T00:00:00\\\",\\\"2026-08-04T00:00:00\\\",\\\"2026-08-05T00:00:00
\\\"],\\\"y\\\":{\\\"dtype\\\":\\\"f8\\\",\\\"bdata\\\":\\\"DElTKk4M
```


Console errors: An iframe which has both allow-scripts and allow-same-origin for its sandbox attribute can escape its sandboxing.


AI Brain
Job Car...
Mainten...
Chat
Teams
Energy (kWh)
300
200
100
0
Jun 14
2026
Jun 28
Jul 12
Jul 26
Aug 9
Date
Inverter 1
Inverter 2
Inverter 3
Inverter 4
Inverter 5
Inverter 6
View Fullscreen
10:03 AM
Are you satisfied with the answer?
Time taken: 27s, token: 0 tokens
connected to Bf Wheeling
How can we help you today?
LEAVE ANY
FEEDBACK
Mistakes in P2Chat are possible. Double-check key
information.

8. Abdali_Mall_Barza

Production • 49.0s

Question: Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well

Error reason: AI indicated failure (matched 'tool error'): Clipping Loss Breakdown (Last 6 Months) - Unable to calculate per-inverter clipping losses due to an internal tool error. • Error: `calculate_clipping_or_curtailment_losses() got an unexpected

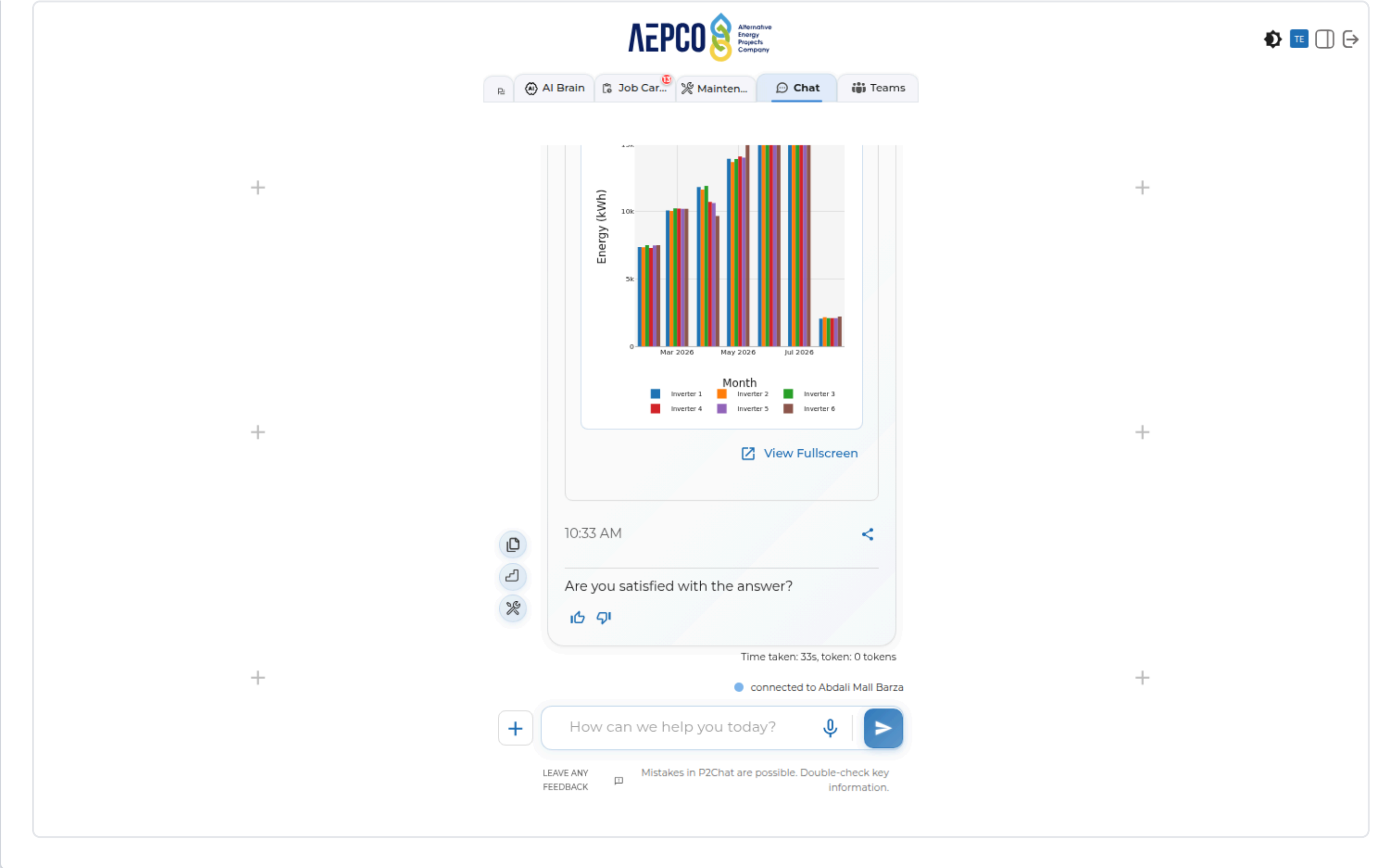
TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "Please figure out how much clipping loss was found in the last 6 months broken by invert
ers, list them down and plot it as well",
    "displayed_response": "Clipping Loss Breakdown (Last 6 Months)\n\n- Unable to calculate per-inverter cl
ipping losses due to an internal tool error.\n  • Error: `calculate_clipping_or_curtailment_losses() got an
unexpected keyword argument 'reactive_col'`\n\nVisualization\n\n- A fallback monthly energy by inverter cha
rt has been prepared (titled \"Monthly Energy by Inverter (Fallback Plot: clipping-loss tool failed)\").",
    "coding_response": [
      "I couldn't calculate the clipping loss breakdown because the clipping-loss tool failed internally fo
r this plant/query.\n\nThe tool error was:\n`calculate_clipping_or_curtailment_losses() got an unexpected k
eyword argument 'reactive_col'`\n\nSo I'm not able to provide trustworthy per-inverter clipping-loss values
for the last 6 months from the available tools right now.",
      {
        "type": "graph",
        "series": [
          "<html>\n<head><meta charset=\"utf-8\" /></head>\n<body>\n    <div>                        <scrip
t type=\"text/javascript\">window.PlotlyConfig = {MathJaxConfig: 'local'};</script>\n        <script charse
t=\"utf-8\" src=\"https://cdn.plot.ly/plotly-3.0.1.min.js\"></script>                <div id=\"07e5ecf4-440
8-466e-98d9-e40e5b46ad2b\" class=\"plotly-graph-div\" style=\"height:100%; width:100%;\"></div>
<script type=\"text/javascript\">                window.PLOTLYENV=window.PLOTLYENV || {};
if (document.getElementById(\"07e5ecf4-4408-466e-98d9-e40e5b46ad2b\")) {                    Plotly.newPlot(
\"07e5ecf4-4408-466e-98d9-e40e5b46ad2b\",                        [{\"hovertemplate\":\"%{fullData.name}: %
{y:.2f} kWh\\u003cextra\\u003e\\u003c\\u002fextra\\u003e\",\"name\":\"Inverter 1\",\"x\":[\"2026-02-01T00:0
0:00\",\"2026-03-01T00:00:00\",\"2026-04-01T00:00:00\",\"2026-05-01T00:00:00\",\"2026-06-01T00:00:00\",\"20
26-07-01T00:00:00\",\"2026-08-01T00:00:00\"],\"y\":{\"dtype\":\"f8\",\"bdata\":\"Gdv5fljEvEDlXc\\u002fqG7TD
QKnEPKsdGcdAC9zDS9cry0DsfdCg4BLOQOGBV0smS85AiTBrT+QkoEA=\"},\"type\":\"bar\"},{\"hovertemplate\":\"%{fullDa
ta.name}: %{y:.2f} kWh\\u003cextra\\u003e\\u003c\\u002fextra\\u003e\",\"name\":\"Inverter 2\",\"x\":[\"2026
-02-01T00:00:00\",\"2026-03-01T00:00:00\",\"2026-04-01T00:00:00\",\"2026-05-01T00:00:00\",\"2026-06-01T00:0
0:00\",\"2026-07-01T00:00:00\",\"2026-08-01T00:00:00\"],\"y\":{\"dtype\":\"f8\",\"bdata\":\"8CXZ\\u002fd63v
EC0SdMgh5\\u002fDQMXXeMPTuMZANAnnQLS2ykDCjae4uz\\u002fQQFrE\\u002fK8cONBAyQ0LyIHnoEA=\"},\"type\":\"bar\"},
{\"hovertemplate\":\"%{fullData.name}: %{y:.2f} kWh\\u003cextra\\u003e\\u003c\\u002fextra\\u003e\",\"name
\":\"Inverter 3\",\"x\":[\"2026-02-01T00:00:00\",\"2026-03-01T00:00:00\",\"2026-04-01T00:00:00\",\"2026-05-
01T00:00:00\",\"2026-06-01T00:00:00\",\"2026-07-01T00:00:00\",\"2026-08-01T00:00:00\"],\"y\":{\"dtype\":\"f
8\",\"bdata\":\"bdmJpXFQvUAGk6DSWPzDQET4WdIAQcdAwIGtkmwqy0A5qJ5fcV\\u002fOQNUvz128qc5AIepRy41ooEA=\"},\"typ
e\":\"bar\"},{\"hovertemplate\":\"%{fullData.name}: %{y:.2f} kWh\\u003cextra\\u003e\\u003c\\u002fextra\\u00
3e\",\"name\":\"Inverter 4\",\"x\":[\"2026-02-01T00:00:00\",\"2026-03-01T00:00:00\",\"2026-04-01T00:00:00
\",\"2026-05-01T00:00:00\",\"2026-06-01T00:00:00\",\"2026-07-01T00:00:00\",\"2026-08-01T00:00:00\"],\"y\":
{\"dtype\":\"f8\",\"bdata\":\"z3BiNQuIvEAKUmsHCPvDQN7BYueh8sRAz7SetMSGy0Cl3JDZEEjOQBrLb77Gjs5AV\\u002ft0PMJ
hoEA=\"},\"type\":\"bar\"},{\"hovertemplate\":\"%{fullData.name}: %{y:.2f} kWh\\u003cextra\\u003e\\u003c\\u
002fextra\\u003e\",\"name\":\"Inverter 5\",\"x\":[\"2026-02-01T00:00:00\",\"2026-03-01T00:00:00\",\"2026-04
-01T00:00:00\",\"2026-05-01T00:00:00\",\"2026-06-01T00:00:00\",\"2026-07-01T00:00:00\",\"2026-08-01T00:00:0
0\"],\"y\":{\"dtype\":\"f8\",\"bdata\":\"Pq7ezhBFvUAxA8WiE+zDQOlcuXEoxsRAPy9zp6xXy0DS5wyfI1DOQJ0jw5dyjs5AD2
4f3\\u002fNOoEA=\"},\"typ
```


Console errors: An iframe which has both allow-scripts and allow-same-origin for its sandbox attribute can escape its sandboxing.


AEPCO
Alternative Energy Projects Company
AI Brain
Job Car...
Mainten...
Chat
Teams
Energy (kWh)
10k
5k
0
Mar 2026
May 2026
Jul 2026
Month
Inverter 1
Inverter 2
Inverter 3
Inverter 4
Inverter 5
Inverter 6
View Fullscreen
10:33 AM
Are you satisfied with the answer?
Time taken: 33s, token: 0 tokens
connected to Abdali Mall Barza
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

9. Abdali_Mall_Barza

Production • 51.2s

Question: Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

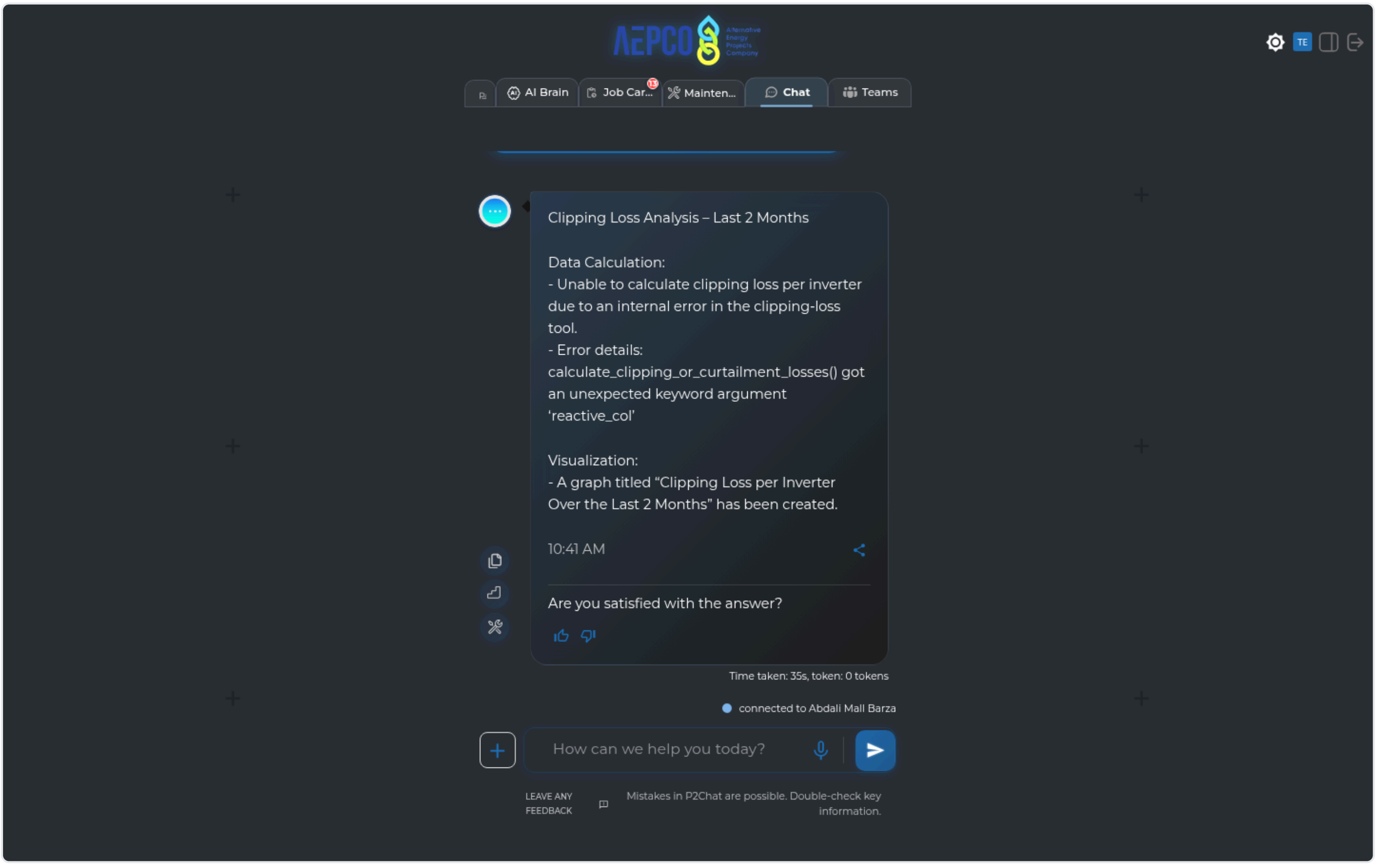
Error reason: AI indicated failure (matched 'internal error'): Clipping Loss Analysis – Last 2 Months Data Calculation: - Unable to calculate clipping loss per inverter due to an internal error in the clipping-loss tool. - Error details: calculate_clipping_or_cu

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "Please figure out how much clipping loss was found in the last 2 months broken by invert
ers, list them down and plot it as well",
    "displayed_response": "Clipping Loss Analysis – Last 2 Months\n\nData Calculation:\n- Unable to calcula
te clipping loss per inverter due to an internal error in the clipping-loss tool.\n- Error details: calcula
te_clipping_or_curtailment_losses() got an unexpected keyword argument 'reactive_col'\n\nVisualization:\n-
A graph titled "Clipping Loss per Inverter Over the Last 2 Months" has been created.",
    "coding_response": [
      "I couldn't calculate clipping loss per inverter for the last 2 months because the clipping-loss tool
failed internally.\n\nThe tool returned this error:\n`calculate_clipping_or_curtailment_losses() got an une
xpected keyword argument 'reactive_col'`\n\nSo I can't provide reliable clipping-loss values from the avail
able tooling right now.",
      {
        "type": "graph",
        "series": [
          "",
          "",
          "",
          ""
        ],
        "graphTitle": "Clipping Loss per Inverter Over the Last 2 Months",
        "yAxisTitle": "Clipping Loss (kWh)"
      }
    ],
    "used_agents": [
      "data_agent",
      "graph_agent"
    ],
    "used_tools": [
      "functions.calculate_clipping_or_curtailment_losses_ag",
      "functions.inspect_xpool",
      "functions.total_energy_func_ag"
    ],
    "urls": [],
    "steps": "1. I filter the data to include only records from June 5, 2026, to August 5, 2026, including
the last day.\n\n2. I find all relevant inverter power columns in the dataset, making sure to leave out any
columns related to reactive power or flags.\n\n3. I check that the required inverter power columns are pres
ent in the data.\n\n4. I clean the plane of array (POA) irradiance data by filling in any missing values us
ing a method that accounts for daily sunlight patterns.\n\n5. For each inverter, I calculate power losses b
y finding the difference between the measured power and the inverter's rated AC power during periods when l
osses are detected.\n   - I identify these loss periods (clipping events) using both a machine learning mod
el (rdtools XGBoost clip filter) and a rule that flags power readings above 85% of the inverter's rated AC
power.\n\n6. I convert the detected power losses into energy lost (in kWh) by multiplying the loss by the t
ime interval (0.0833 hours).\n\n7. I group the energy loss values by the chosen time period (such as day, m
onth, or year) for each inverter.\n\n8. I sum up the grouped values to get a total loss for each inverter o
ver the selected period.\n\n9. I organize the results into a table, with each inverter as a column and each
time period as a row.\n\n10. I return the results as a dictionary, either broken down by each time period o
r as a single total, depending on the settings."
  }
}
```



LLM Reasoning Issues LLM REASONING (6 failures)

1. Noor_AI_aqaba

Production • 49.0s

Question: When was the last time the BOM data showed up

Error reason: AI indicated failure (matched 'an error'): Sorry, there was an error processing your request. 08:03 AM Do you want to report this error? YES NO Time taken: s, token: tokens connected to Noor AI Aqaba LEAVE ANY FEEDBACK Mistakes in P2Cha

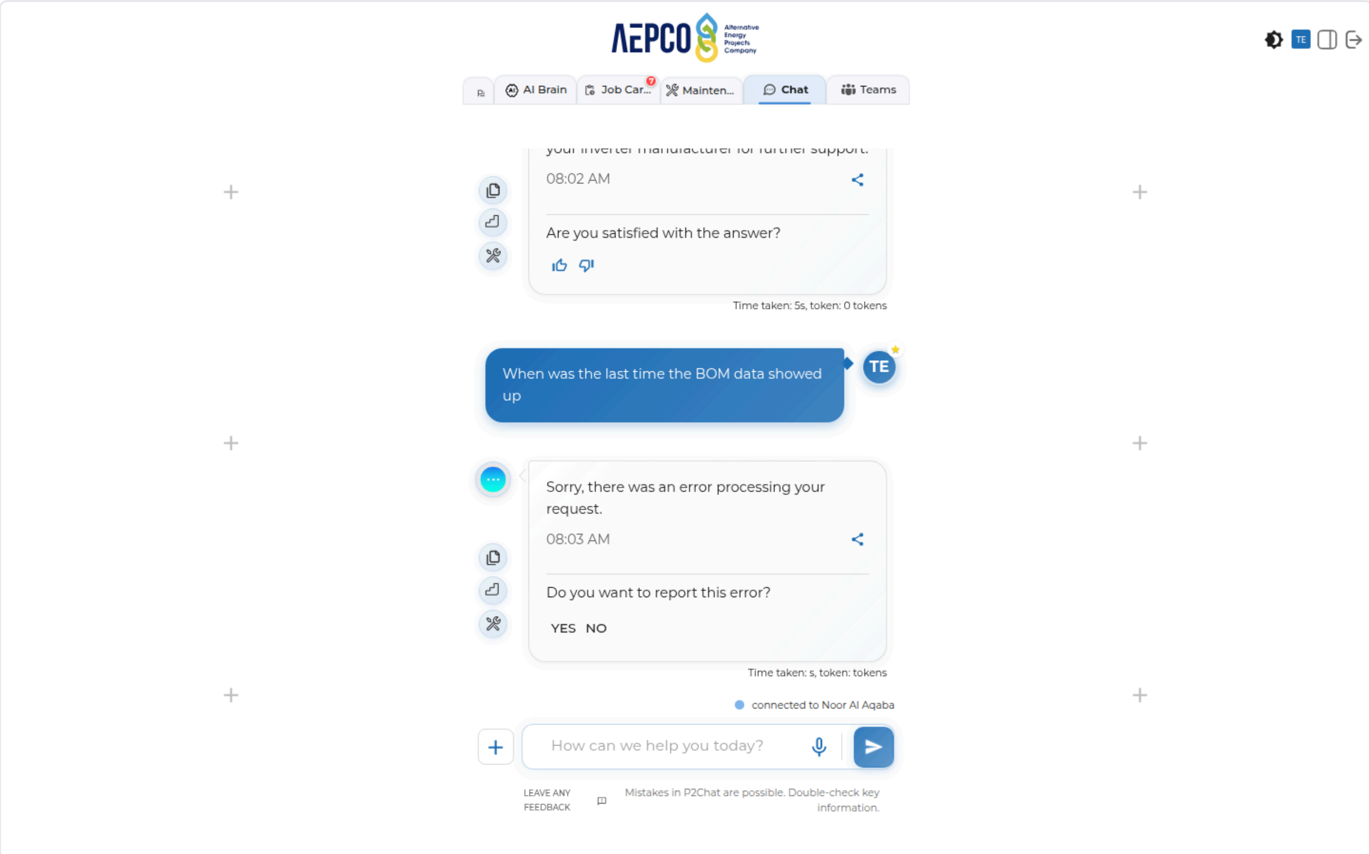
TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 500`

Raw API response:


```
{
  "error": "An unexpected error occurred. Please try again later."
}
```


Console errors: Failed to load resource: the server responded with a status of 500 (Internal Server Error)



2. MOF

Production • 49.0s

Question: When was the last time the BOM data showed up

Error reason: AI indicated failure (matched 'an error'): Sorry, there was an error processing your request. 08:22 AM Do you want to report this error? YES NO Time taken: s, token: tokens connected to Mof LEAVE ANY FEEDBACK Mistakes in P2Chat are poss

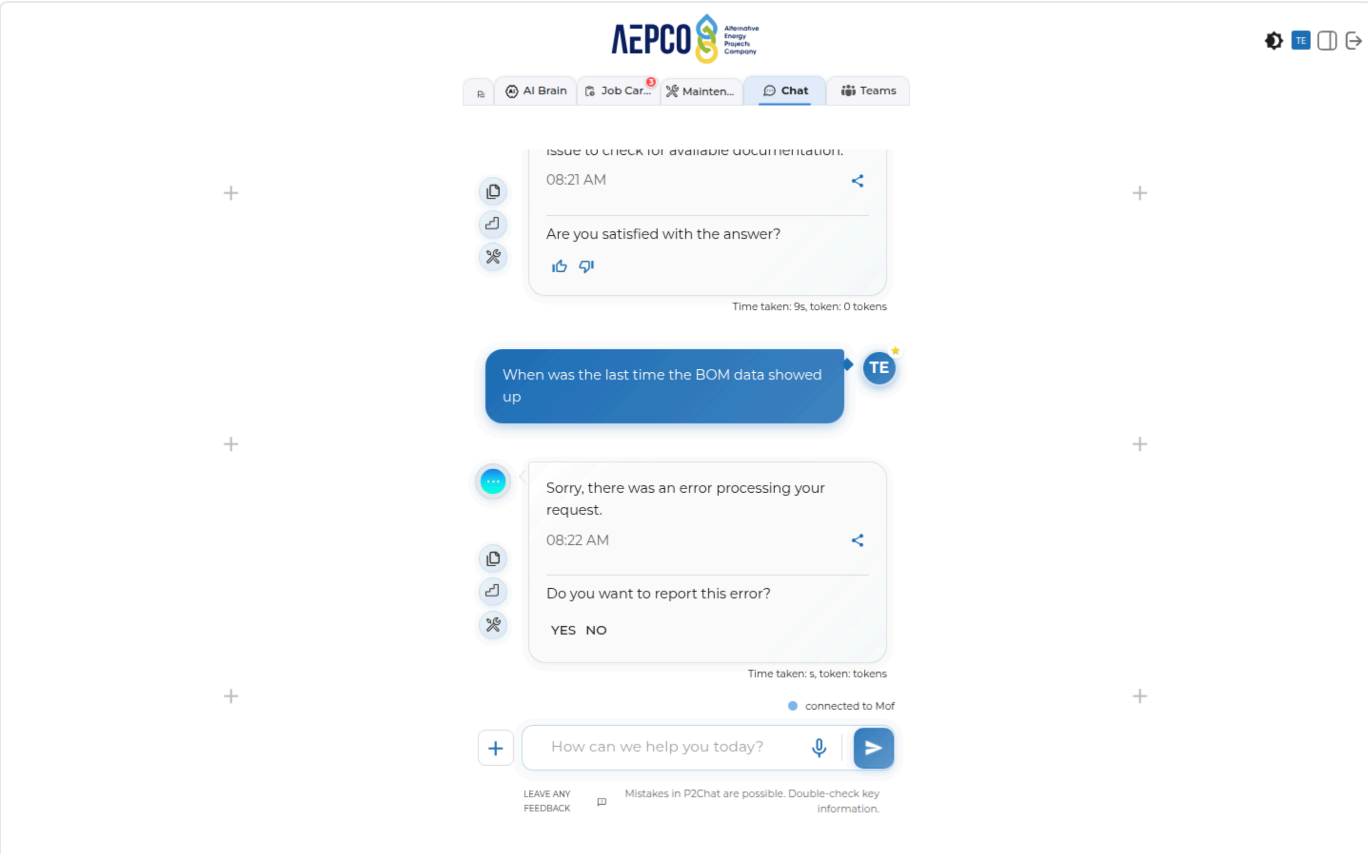
TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 500`

Raw API response:


```
{
  "error": "An unexpected error occurred. Please try again later."
}
```


Console errors: Failed to load resource: the server responded with a status of 500 (Internal Server Error)



3. GID

Production • 23.2s

Question: Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times

Error reason: AI indicated failure (matched 'encountered an issue'): Sorry, we encountered an issue processing your request. Our team has been notified and will investigate. 09:27 AM Do you want to report this error? YES NO Time taken: s, token: tokens connected t

TECHNICAL EVIDENCE

Exception	`AttributeError: 'NoneType' object has no attribute 'rule_code'`
Agent / Tool	`data_agent / extract_timeseries_data_ag`
Endpoint	`/testuser1/ai_agent_chat`
Location	`agent_graph.py:916`

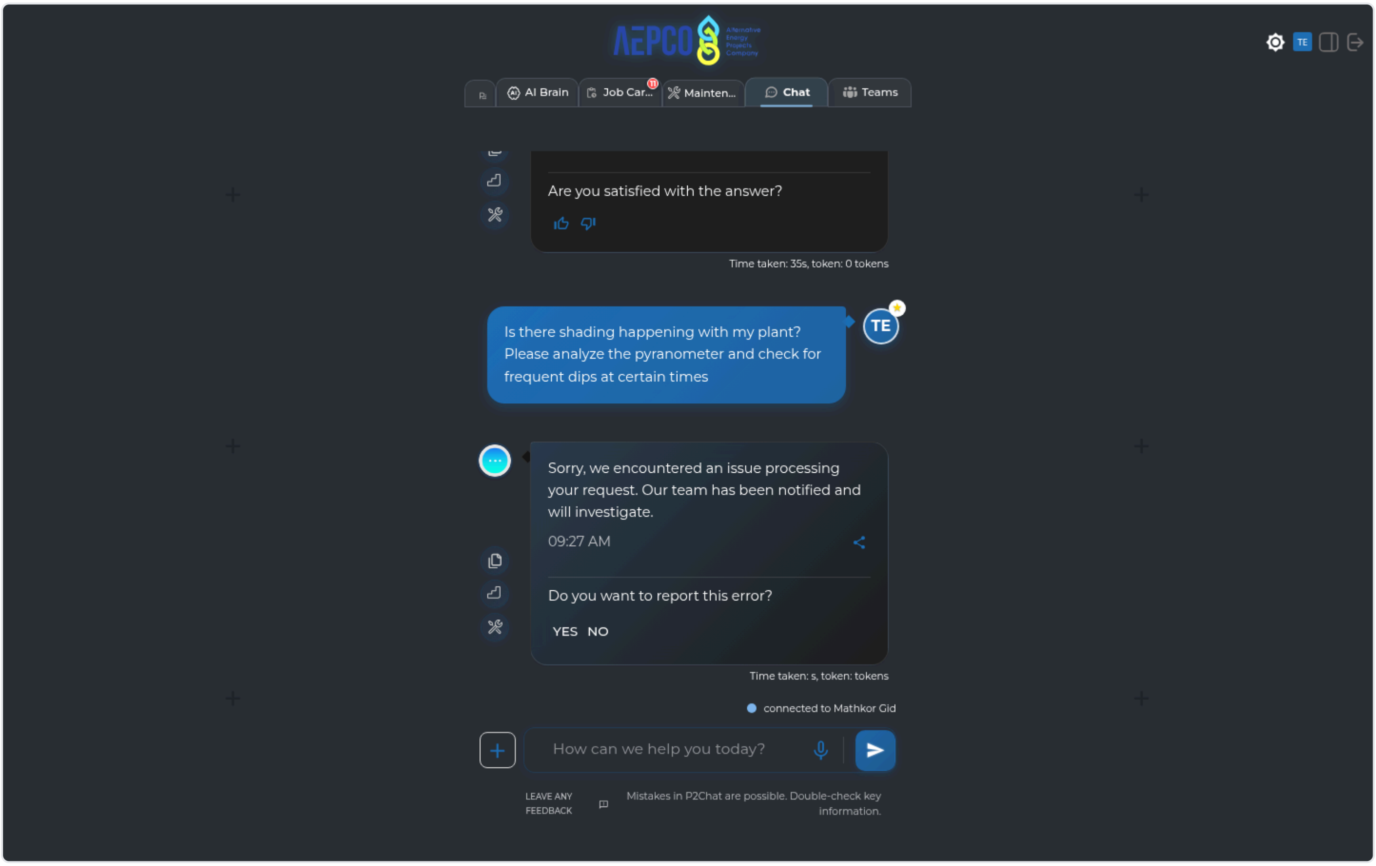
API call: `GET /testuser2/ai_agent_chat → 400`

Raw API response:


```
{
  "displayed_response": "Sorry, we encountered an issue processing your request. Our team has been notified and will investigate.",
  "coding_response": [
    "==================== ERROR SUMMARY ====================\nAPI ENDPOINT: /testuser1/ai_agent_chat\nAGENT: data_agent\nTOOL: extract_timeseries_data_ag\nERROR TYPE: AttributeError\nERROR MESSAGE: 'NoneType' object has no attribute 'rule_code'\nLOCATION: agent_graph.py:916\n======================================================="
  ]
}
```


```
==================== ERROR SUMMARY ====================
API ENDPOINT: /testuser1/ai_agent_chat
AGENT: data_agent
TOOL: extract_timeseries_data_ag
ERROR TYPE: AttributeError
ERROR MESSAGE: 'NoneType' object has no attribute 'rule_code'
LOCATION: agent_graph.py:916
=======================================================
```

Console errors: Failed to load resource: the server responded with a status of 400 (Bad Request)


AI Brain
Job Car...
Mainten...
Chat
Teams
Are you satisfied with the answer?
Time taken: 35s, token: 0 tokens
TE
Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times
Sorry, we encountered an issue processing your request. Our team has been notified and will investigate.
09:27 AM
Do you want to report this error?
YES NO
Time taken: s, token: tokens
connected to Mathkor Gid
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

4. Abdali_Mall_Netmetering

Production • 43.1s

Question: Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times

Error reason: AI indicated failure (matched 'encountered an issue'): Sorry, we encountered an issue processing your request. Our team has been notified and will investigate. 09:46 AM Do you want to report this error? YES NO Time taken: s, token: tokens connected t

TECHNICAL EVIDENCE

Exception	`PydanticSerializationError: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>`
Agent / Tool	`data_agent / unknown`
Endpoint	`/testuser1/ai_agent_chat`
Location	`agent_graph.py:916`

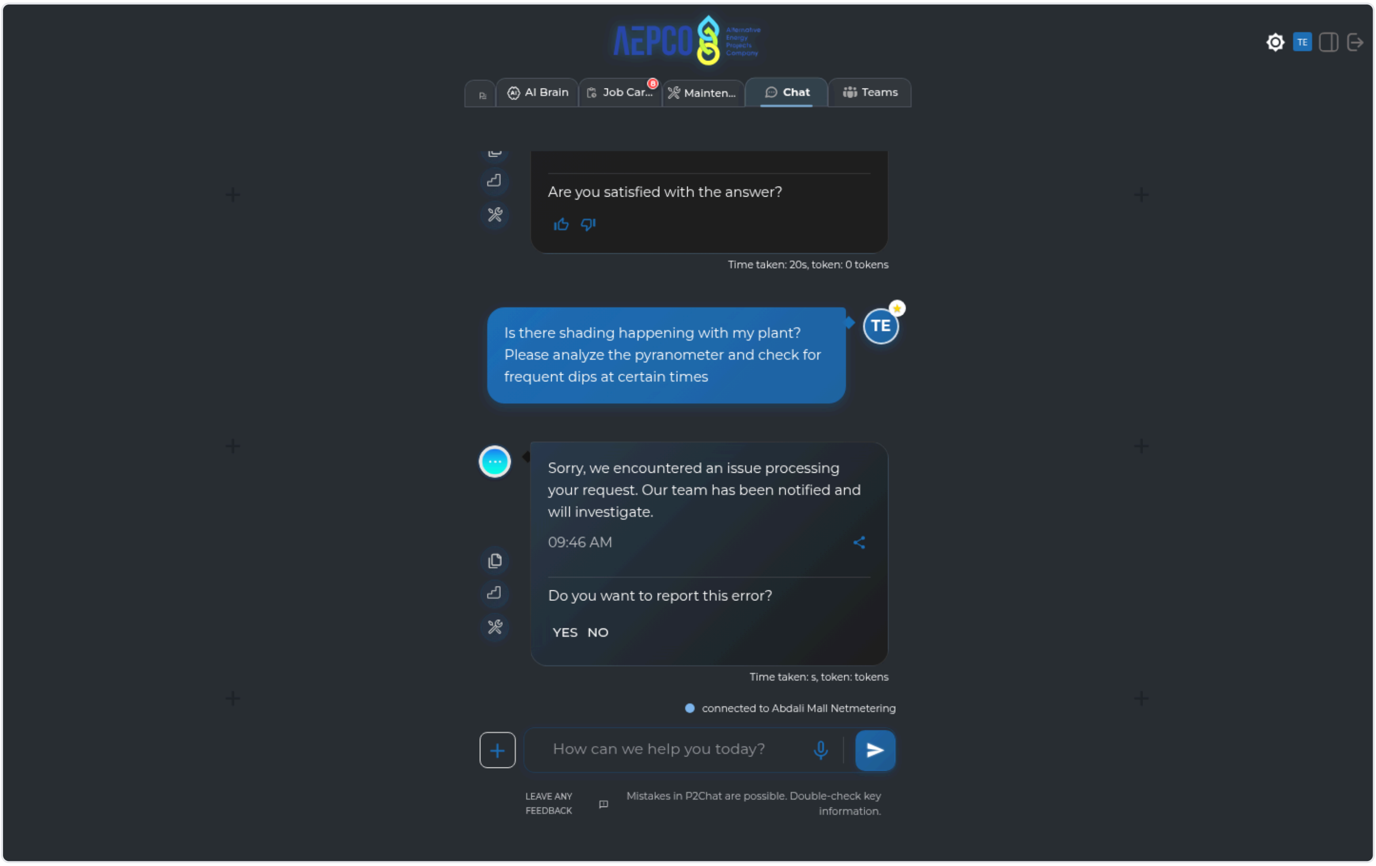
API call: `GET /testuser2/ai_agent_chat → 400`

Raw API response:


```
{
  "displayed_response": "Sorry, we encountered an issue processing your request. Our team has been notified and will investigate.",
  "coding_response": [
    "=================== ERROR SUMMARY ===================\nAPI ENDPOINT: /testuser1/ai_agent_chat\nAGENT: data_agent\nTOOL: unknown\nERROR TYPE: PydanticSerializationError\nERROR MESSAGE: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>\nLOCATION: agent_graph.py:916\n======================================================="
  ]
}
```


```
=================== ERROR SUMMARY ===================
API ENDPOINT: /testuser1/ai_agent_chat
AGENT: data_agent
TOOL: unknown
ERROR TYPE: PydanticSerializationError
ERROR MESSAGE: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>
LOCATION: agent_graph.py:916
=======================================================
```

Console errors: Failed to load resource: the server responded with a status of 400 (Bad Request)


AI Brain
Job Car...
Mainten...
Chat
Teams
Are you satisfied with the answer?
Time taken: 20s, token: 0 tokens
TE
Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times
Sorry, we encountered an issue processing your request. Our team has been notified and will investigate.
09:46 AM
Do you want to report this error?
YES NO
Time taken: s, token: tokens
connected to Abdali Mall Netmetering
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

5. BF_Wheeling

Production • 21.6s

Question: How can I shut down my inverters?

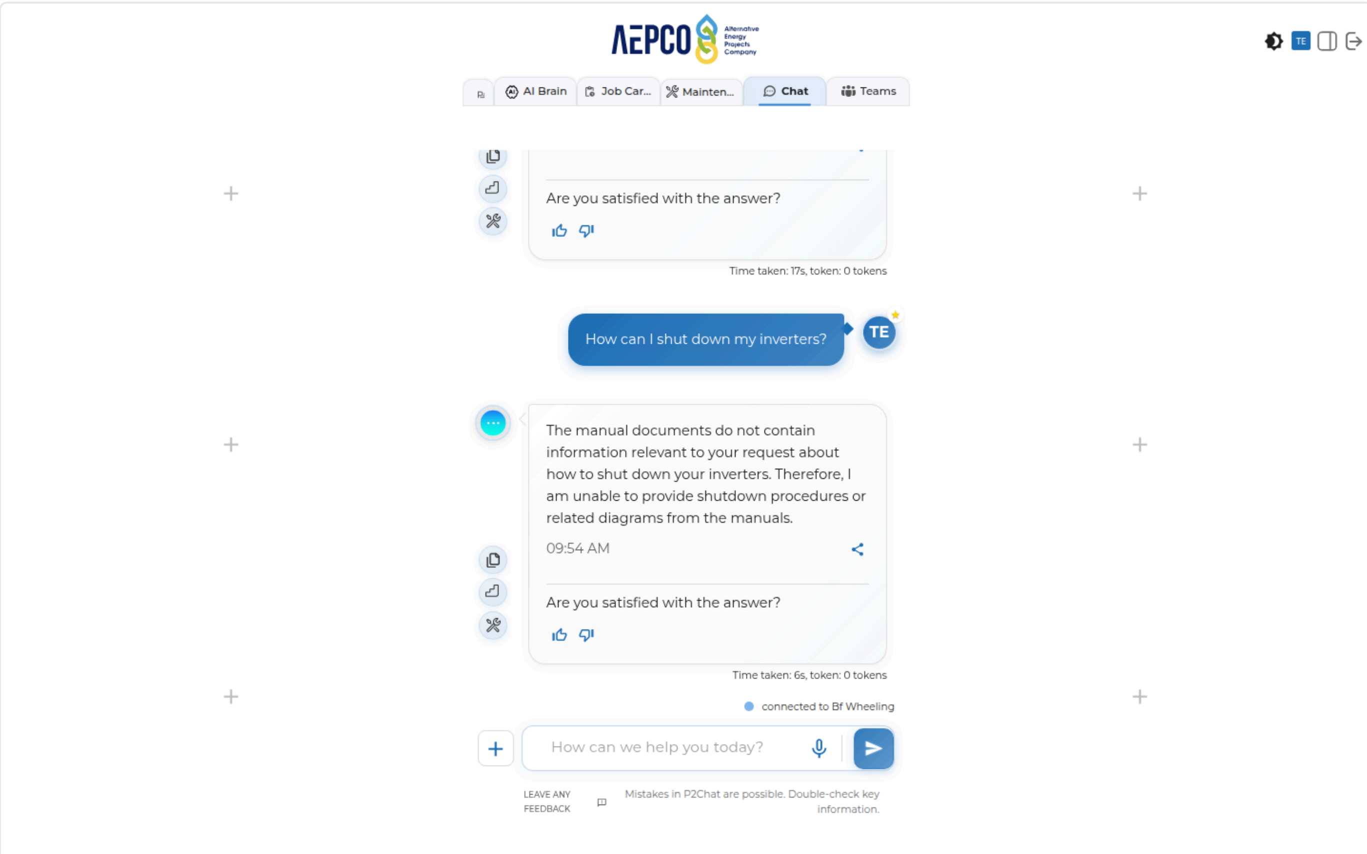
Error reason: AI indicated failure (matched 'unable to provide'): The manual documents do not contain information relevant to your request about how to shut down your inverters. Therefore, I am unable to provide shutdown procedures or related diagrams from the manua

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "How can I shut down my inverters?",
    "displayed_response": "The manual documents do not contain information relevant to your request about h
ow to shut down your inverters. Therefore, I am unable to provide shutdown procedures or related diagrams f
rom the manuals.",
    "coding_response": [
      "The manual documents do not contain information relevant to your request about how to shut down your
inverters. Therefore, I am unable to provide shutdown procedures or related diagrams from the manuals."
    ],
    "used_agents": [
      "manual_agent"
    ],
    "used_tools": [
      "functions.manual_rag_tool"
    ],
    "urls": [],
    "steps": []
  }
}
```

6. Abdali_Mall_Lahoun

Production • 38.5s

Question: Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times

Error reason: AI indicated failure (matched 'encountered an issue'): Sorry, we encountered an issue processing your request. Our team has been notified and will investigate. 10:22 AM Do you want to report this error? YES NO Time taken: s, token: tokens connected t

TECHNICAL EVIDENCE

Exception	`PydanticSerializationError: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>`
Agent / Tool	`data_agent / unknown`
Endpoint	`/testuser1/ai_agent_chat`
Location	`agent_graph.py:916`

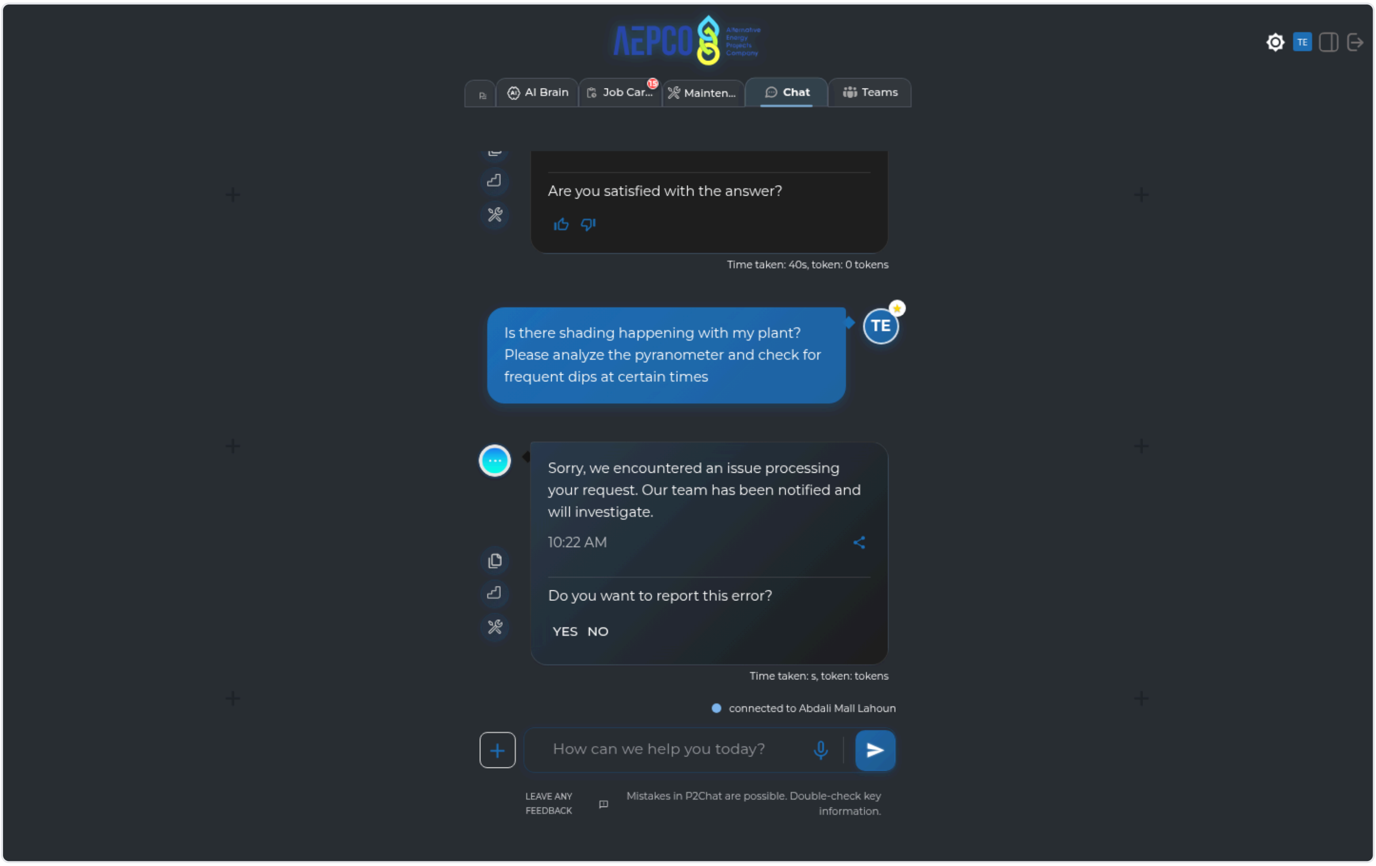
API call: `GET /testuser2/ai_agent_chat → 400`

Raw API response:


```
{
  "displayed_response": "Sorry, we encountered an issue processing your request. Our team has been notified and will investigate.",
  "coding_response": [
    "=================== ERROR SUMMARY ===================\nAPI ENDPOINT: /testuser1/ai_agent_chat\nAGENT: data_agent\nTOOL: unknown\nERROR TYPE: PydanticSerializationError\nERROR MESSAGE: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>\nLOCATION: agent_graph.py:916\n======================================================="
  ]
}
```


```
=================== ERROR SUMMARY ===================
API ENDPOINT: /testuser1/ai_agent_chat
AGENT: data_agent
TOOL: unknown
ERROR TYPE: PydanticSerializationError
ERROR MESSAGE: Unable to serialize unknown type: <class 'pandas.core.frame.DataFrame'>
LOCATION: agent_graph.py:916
=======================================================
```

Console errors: Failed to load resource: the server responded with a status of 400 (Bad Request)



Data Availability Issues DATA AVAILABILITY (4 failures)

1. KUPCO_HQ_DAJEEJ

Production • 43.0s

Question: Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well

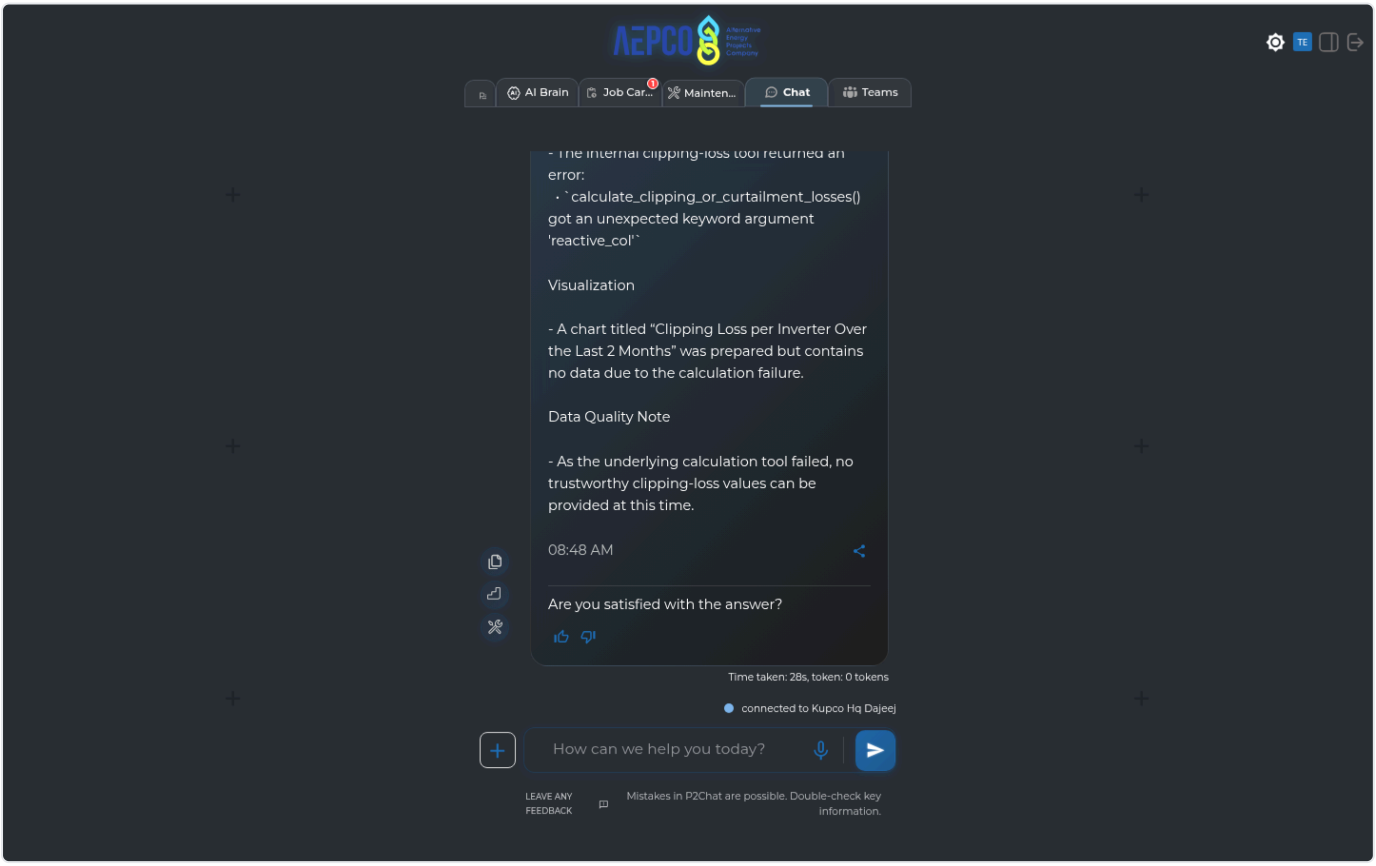
Error reason: AI indicated failure (matched 'an error'): Data Availability - Clipping loss breakdown by inverter for the last 2 months could not be calculated. - The internal clipping-loss tool returned an error: • `calculate_clipping_or_curtailment_loss

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "Please figure out how much clipping loss was found in the last 2 months broken by invert
ers, list them down and plot it as well",
    "displayed_response": "Data Availability\n\n- Clipping loss breakdown by inverter for the last 2 months
could not be calculated.\n- The internal clipping-loss tool returned an error:\n  • `calculate_clipping_or_
curtailment_losses() got an unexpected keyword argument 'reactive_col'`\n\nVisualization\n\n- A chart title
d “Clipping Loss per Inverter Over the Last 2 Months” was prepared but contains no data due to the calculat
ion failure.\n\nData Quality Note\n\n- As the underlying calculation tool failed, no trustworthy clipping-l
oss values can be provided at this time.",
    "coding_response": [
      "I couldn’t calculate clipping loss for the last 2 months because the clipping-loss tool failed inter
nally on this dataset/tool version.\n\nThe tool returned this error:\n- `calculate_clipping_or_curtailment_
losses() got an unexpected keyword argument 'reactive_col'`\n\nSo I can’t provide a trustworthy clipping-lo
ss breakdown by inverter from the available tools right now.",
      {
        "type": "graph",
        "series": [
          "",
          "",
          "",
          ""
        ],
        "graphTitle": "Clipping Loss per Inverter Over the Last 2 Months",
        "yAxisTitle": "Clipping Loss (kWh)"
      }
    ],
    "used_agents": [
      "data_agent",
      "graph_agent"
    ],
    "used_tools": [
      "functions.inspect_xpool",
      "functions.total_energy_func_ag",
      "functions.calculate_clipping_or_curtailment_losses_ag"
    ],
    "urls": [],
    "steps": "1. I filter the data to include only records from June 5, 2026, to August 5, 2026, including
the last day.\n2. I find all relevant inverter power columns in the dataset, making sure to leave out any c
olumns related to reactive power or flags.\n3. I check that the required power columns are actually present
in the data.\n4. I clean up the plane of array (POA) irradiance data by filling in any missing values using
a method that accounts for daily sunlight patterns.\n5. For each inverter, I calculate power losses by find
ing the absolute difference between the measured power and the inverter’s rated AC power during periods whe
n losses are detected.\n   - I identify clipping events using both a machine learning approach (rdtools XGB
oost clip filter) and a rule that flags power readings above 85% of the inverter’s rated AC power.\n6. I co
nvert these power losses into energy losses (in kWh) by multiplying by the time interval of 0.0833 hours.\n
7. I sum up the energy losses for each inverter, grouping the results by month (or by day, month, or year,
depending on the setting).\n8. I provide a detailed breakdown of the loss values for each inverter and each
time period.\n9. I organize the results into a DataFrame, with each inverter as a column and each time peri
od as a row.\n10. I return the results as a dictionary, where each key is a time period (or a single total
row), depending on the chosen parameters."
  }
}
```



AI Brain
Job Car...
Mainten...
Chat
Teams
- The internal clipping-loss tool returned an error:
• `calculate_clipping_or_curtailment_losses() got an unexpected keyword argument 'reactive_col'`
Visualization
- A chart titled "Clipping Loss per Inverter Over the Last 2 Months" was prepared but contains no data due to the calculation failure.
Data Quality Note
- As the underlying calculation tool failed, no trustworthy clipping-loss values can be provided at this time.
08:48 AM
Are you satisfied with the answer?
Time taken: 28s, token: 0 tokens
connected to Kupco Hq Dajeej
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

2. Galleria_Mall

Production • 33.9s

Question: How did my plant perform yesterday?

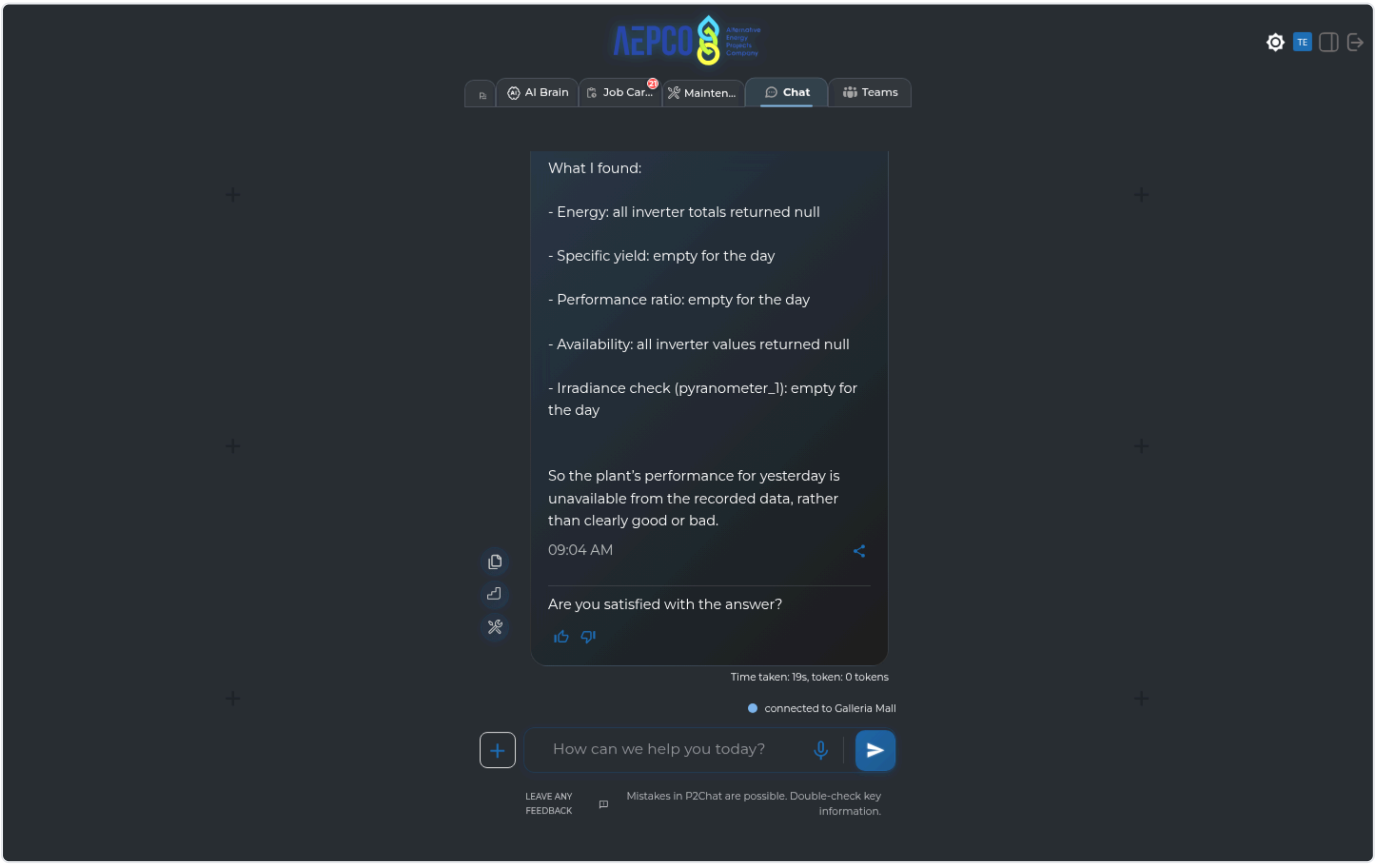
Error reason: AI indicated failure (matched 'no data available'): I can't assess yesterday's performance because there is no data available for 2026-08-04 in the database. What I found: - Energy: all inverter totals returned null - Specific yield: empty for t

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "How did my plant perform yesterday?",
    "displayed_response": "I can't assess yesterday's performance because there is no data available for 20
26-08-04 in the database.\n\nWhat I found:\n- Energy: all inverter totals returned null\n- Specific yield:
empty for the day\n- Performance ratio: empty for the day\n- Availability: all inverter values returned nul
l\n- Irradiance check (pyranometer_1): empty for the day\n\nSo the plant's performance for yesterday is una
vailable from the recorded data, rather than clearly good or bad.",
    "coding_response": [
      "I can't assess yesterday's performance because there is no data available for 2026-08-04 in the data
base.\n\nWhat I found:\n- Energy: all inverter totals returned null\n- Specific yield: empty for the day\n-
Performance ratio: empty for the day\n- Availability: all inverter values returned null\n- Irradiance check
(pyranometer_1): empty for the day\n\nSo the plant's performance for yesterday is unavailable from the reco
rded data, rather than clearly good or bad."
    ],
    "used_agents": [
      "data_agent"
    ],
    "used_tools": [
      "functions.extract_timeseries_data_ag",
      "multi_tool_use.parallel"
    ],
    "urls": [],
    "steps": "1. I select data from 2026-08-04 to 2026-08-04, including the end date.\n2. I find all invert
er power columns and their DC rated power values using the plant metadata.\n3. I set calculation parameters
like the irradiance threshold (usually 50 W/m²), power threshold (usually 0.1 kW), and the data interval (t
ypically 5 minutes).\n4. For each inverter, I filter the data to only include times when irradiance is abov
e the threshold, then count the total useful intervals (T_useful) and the downtime intervals (T_down).\n5.
I convert these interval counts into hours based on the data interval.\n6. I calculate the Technical Availa
bility for each inverter using the formula: (T_useful - T_down) / T_useful * 100.\n7. I return a dictionary
showing the availability percentage for each inverter.\n\n1. I select data from 2026-08-04 to 2026-08-04, i
ncluding the end date.\n2. I identify all inverter power columns in the dataset, excluding columns related
to reactive power and flags.\n3. I clean the data by removing or marking invalid measurements in each inver
ter's power column.\n4. For each inverter, I calculate energy by multiplying power by the time interval (0.
0833 hours) and adjust for units (for example, divide by 1000 if the data is in Watts).\n5. I include all d
ays in the calculation for each inverter.\n6. I sum up all energy values in the selected period to get a to
tal energy value for each inverter.\n7. I organize the results into a DataFrame with inverters as columns a
nd aggregation periods as the index.\n8. I return the results as a dictionary, either by aggregation period
or as a single total row, depending on the settings.\n\n1. I select data from 2026-08-04 to 2026-08-04, inc
luding the end date.\n2. I identify the relevant columns for power, irradiance, and temperature.\n3. I clea
n the data by removing or flagging invalid values in the power and irradiance columns.\n4. I calculate the
actual energy produced by multiplying power by the time interval (0.0833 hours) and adjusting for units.\n
5. I calculate the expected energy without applying temperature correction.\n6. If the system is bifacial,
I apply extra correction factors based on metadata like ground albedo, area normalization, and shading loss
es.\n7. I group actual and expected energy by daily intervals.\n8. I calculate the performance ratio for ea
ch interval, applying bifacial corrections if needed.\n9. I return the average performance ratio across all
intervals.\n10. I cap the performance ratio values at 1.3 to avoid unrealistic results.\n\n1. I select data
from 2026-08-04 to 2026-08-04, including the end date.\n2. I find all inverter power columns and their DC r
ated power values using the plant metadata.\n3. I clean the data by removing
```


AI Brain
Job Car...
Mainten...
Chat
Teams
What I found:
- Energy: all inverter totals returned null
- Specific yield: empty for the day
- Performance ratio: empty for the day
- Availability: all inverter values returned null
- Irradiance check (pyranometer_1): empty for the day
So the plant's performance for yesterday is unavailable from the recorded data, rather than clearly good or bad.
09:04 AM
Are you satisfied with the answer?
Time taken: 19s, token: 0 tokens
connected to Galleria Mall
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

3. BF_Wheeling

Production • 52.3s

Question: Please generate last months report

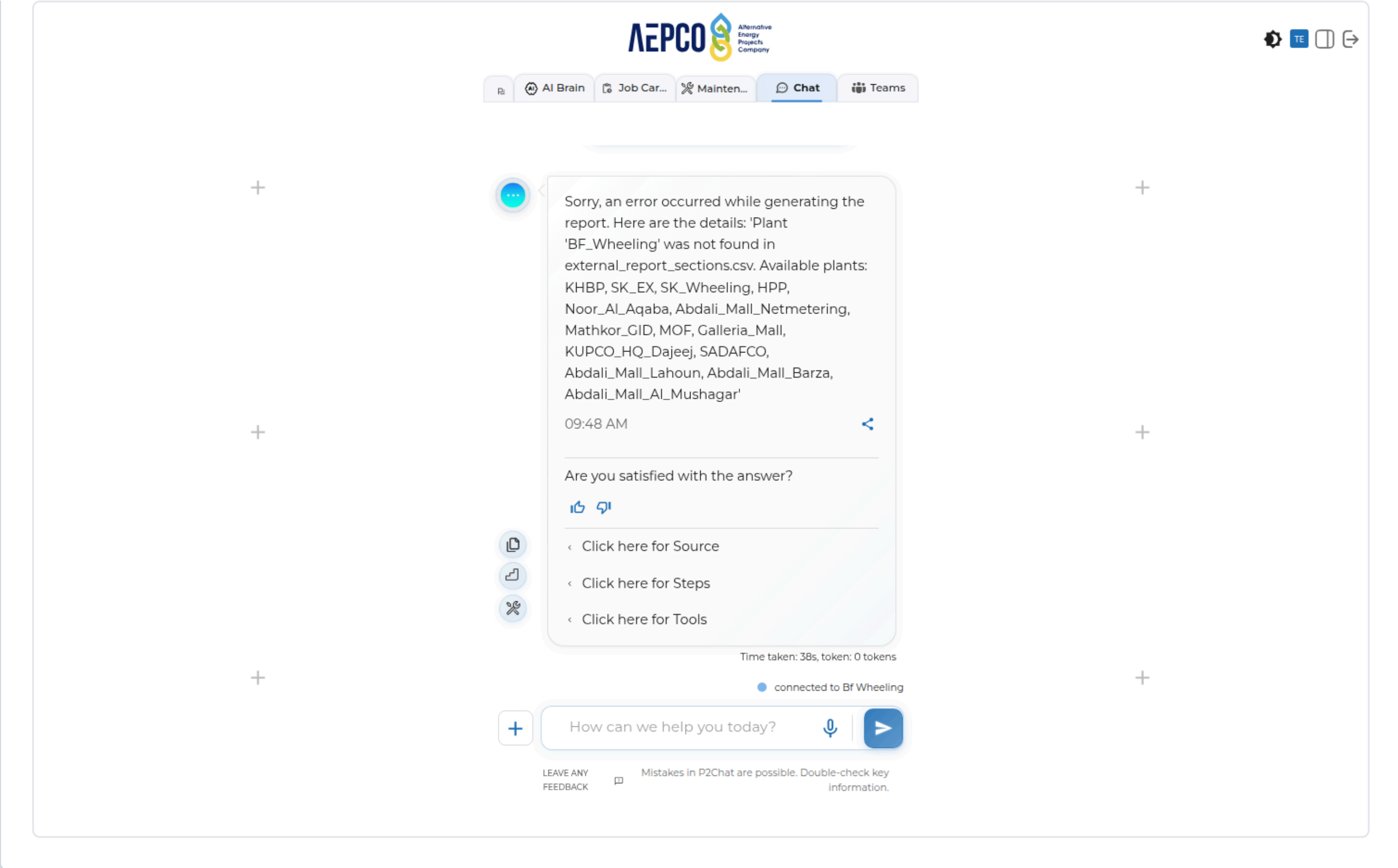
Error reason: AI indicated failure (matched 'error occurred'): Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_Al

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "Please generate last months report",
    "displayed_response": "Sorry, an error occurred while generating the report. Here are the details: 'Pla
nt 'BF_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling,
HPP, Noor_Al_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abd
ali_Mall_Lahoun, Abdali_Mall_Barza, Abdali_Mall_Al_Mushagar'",
    "coding_response": [
      [
        {
          "answer": "Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF
_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP,
Noor_Al_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_M
all_Lahoun, Abdali_Mall_Barza, Abdali_Mall_Al_Mushagar'",
          "action": []
        }
      ]
    ],
    "used_agents": [
      "report_agent"
    ],
    "used_tools": [
      "agent_external_report_tool"
    ],
    "urls": [],
    "steps": []
  }
}
```


AI Brain
Job Car...
Mainten...
Chat
Teams
Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_Al_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_Mall_Lahoun, Abdali_Mall_Barza, Abdali_Mall_Al_Mushagar'
09:48 AM
Are you satisfied with the answer?
Click here for Source
Click here for Steps
Click here for Tools
Time taken: 38s, token: 0 tokens
connected to Bf Wheeling
How can we help you today?
LEAVE ANY FEEDBACK
Mistakes in P2Chat are possible. Double-check key information.

4. BF_Wheeling

Production • 44.5s

Question: Please generate December 2025 performance report

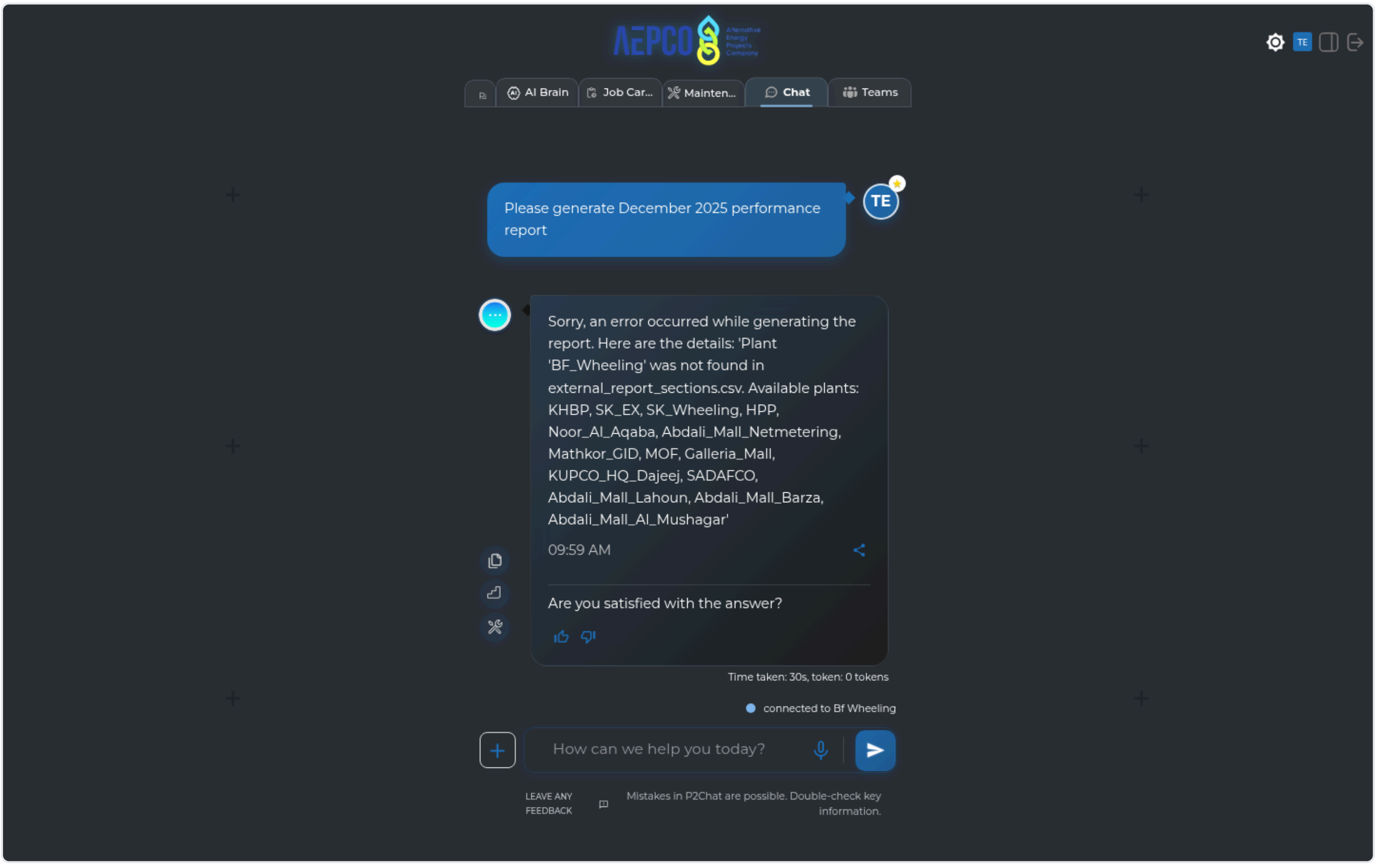
Error reason: AI indicated failure (matched 'error occurred'): Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP, Noor_Al

TECHNICAL EVIDENCE

API call: `GET /testuser2/ai_agent_chat → 200`

Raw API response:


```
{
  "result": {
    "user_input": "Please generate December 2025 performance report",
    "displayed_response": "Sorry, an error occurred while generating the report. Here are the details: 'Pla
nt 'BF_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling,
HPP, Noor_Al_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abd
ali_Mall_Lahoun, Abdali_Mall_Barza, Abdali_Mall_Al_Mushagar'",
    "coding_response": [
      [
        {
          "answer": "Sorry, an error occurred while generating the report. Here are the details: 'Plant 'BF
_Wheeling' was not found in external_report_sections.csv. Available plants: KHBP, SK_EX, SK_Wheeling, HPP,
Noor_Al_Aqaba, Abdali_Mall_Netmetering, Mathkor_GID, MOF, Galleria_Mall, KUPCO_HQ_Dajeej, SADAFCO, Abdali_M
all_Lahoun, Abdali_Mall_Barza, Abdali_Mall_Al_Mushagar'",
          "action": []
        }
      ]
    ],
    "used_agents": [
      "report_agent"
    ],
    "used_tools": [
      "agent_external_report_tool"
    ],
    "urls": [],
    "steps": []
  }
}
```



Skipped (21)

Plant	Question	Env	Skip reason
SAK_1	Please generate last months report	Production	Plant connection failed
SAK_1	Based on pvsyst report, what is the rated power of this plant and array configuration	Production	Plant connection failed
SAK_1	Please plot last 3 days hourly performance ratio for my MPPTs for inverter 2	Production	Plant connection failed
SAK_1	Please plot last 3 days hourly performance ratio for my Strings for inverter 2	Production	Plant connection failed
SAK_1	Please plot the last 4 days hourly performance ratio for all my inverters	Production	Plant connection failed
SAK_1	Please plot my current and voltage in a dual axes plot on the deepest possible granularity level for inverter3	Production	Plant connection failed
SAK_1	Please repeat the first plotting question, but interchange it with Specific yield on 1 axes, and energy production on the other axes on a single dual axes plot (Memory Test)	Production	Plant connection failed
SAK_1	How can I shut down my inverters?	Production	Plant connection failed

Plant	Question	Env	Skip reason
SAK_1	I have a COMMUNICATION_FAULT what do I do to my inverters?	Production	Plant connection failed
SAK_1	When was the last time the BOM data showed up	Production	Plant connection failed
SAK_1	Please figure out how much clipping loss was found in the last 6 months broken by inverters, list them down and plot it as well	Production	Plant connection failed
SAK_1	Was there any 0 production days in the last 7 days?	Production	Plant connection failed
SAK_1	Please change to dark mode	Production	Plant connection failed
SAK_1	Pull up the latest cleaning log	Production	Plant connection failed
SAK_1	Jim cleaned the pyranometer today, can you add it to the cleaning log?	Production	Plant connection failed
SAK_1	Please generate December 2025 performance report	Production	Plant connection failed
SAK_1	How is my data quality for POA and power? How many of them show missing data during daytime or mismatches with satellite prediction?	Production	Plant connection failed
SAK_1	How did my plant perform yesterday?	Production	Plant connection failed
SAK_1	Please find out the daily POA poa summed up as a Wh/m^2, daily for the last 12 months and then calculate the median, mean and kurt	Production	Plant connection failed
SAK_1	Is there shading happening with my plant? Please analyze the pyranometer and check for frequent dips at certain times	Production	Plant connection failed
SAK_1	Please figure out how much clipping loss was found in the last 2 months broken by inverters, list them down and plot it as well	Production	Plant connection failed